



**TRAFFIC VIOLATION DETECTION AND DRIVER SAFETY
SCORING SYSTEM**

Project Final Report
Project ID: 25-26J-330

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April 2026

DECLARATION

I declare that this is my own work, and this report does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any institution of higher education. To the best of my knowledge and belief it does not contain any material previously published or written by another person except where due reference is made in the text.

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ABSTRACT

Road traffic violations remain one of the most preventable causes of accidents, injuries, and fatalities on Sri Lanka's roads. Red light running at signalized intersections and dangerous lane weaving on multi-lane roads are among the most frequently occurring and high-risk violation types, yet both continue to go undetected under manual enforcement systems that cannot scale to the volume and speed of modern urban traffic.

This component report presents the Traffic Violation Detection and Driver Safety Scoring System, developed as one module of the Intelligent Traffic Management System (ITMS, Project ID: 25-26J-330) at the Sri Lanka Institute of Information Technology. The system uses YOLOv8, the latest iteration of the You Only Look Once convolutional neural network family, to detect and classify vehicles in real time from live traffic camera feeds. BoT-SORT (Bootstrapped Object Tracking with SORT) maintains stable vehicle identities across video frames, assigning each vehicle a unique tracking ID that enables the system to monitor individual vehicle trajectories and flag violations without generating duplicate records.

A purpose-built violation rule engine analyses the trajectory and position data produced by the detection and tracking pipeline to classify red light violations and lane weaving events. When a violation is confirmed, the system records it, links it to the responsible driver, and immediately updates that driver's Safety Score with a gamified 100-point daily score that decrements by 15 points for red light violations and 5 points for lane weaving events. Drivers interact with the system through a companion mobile application that displays their score, violation history, and fine payment options in real time. Real-time IoT-based alerts via LEDs and buzzers provide immediate on-site feedback at violation points.

The system is designed for deployment on edge computing hardware, enabling real-time processing with low latency at roadside locations. Evaluation on a simulated Sri Lankan traffic dataset demonstrates mAP@0.5 of 91.4% for vehicle detection and

violation detection F1-scores of 0.942 for red light and 0.877 for lane weaving under normal daytime conditions.

Keywords – *YOLOv8, BoT-SORT, traffic violation detection, lane weaving, red light violation, driver safety score, gamification, mobile application, edge computing, intelligent transportation systems, Sri Lanka.*

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, Prof. Samantha Rajapaksha, for providing consistent and insightful guidance throughout this research project. Prof. Rajapaksha's expertise in intelligent transportation systems and computer vision helped shape both the technical direction of this component and the academic rigor with which it is presented. Without that guidance, the scope and depth of this work would not have been achievable.

I am grateful to the Department of Information Technology at the Sri Lanka Institute of Information Technology for providing the computational infrastructure, laboratory facilities, and academic environment that made this work possible. Access to adequate hardware during the model training and testing phases was critical to the quality of the results.

My sincere appreciation goes to my project teammates in Group 25-26J-330. The broader Intelligent Traffic Management System that this component forms a part of was built through genuine collaboration, and the quality of each component reflects the effort and commitment of the entire group. Working with them reinforced the importance of clear communication, reliable interfaces, and mutual respect in a collaborative engineering project.

I also wish to thank the traffic enforcement officers and transport authority staff whose informal insights into the day-to-day realities of traffic monitoring in Sri Lanka helped ensure that this system addresses genuine practical needs rather than only theoretical ones.

Finally, to my family and friends thank you for the patience, the encouragement, and the support throughout the many long hours this project required. This work is dedicated to you.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BoT-SORT	Bootstrapped Object Tracking with Simple Online and Realtime Tracking
CNN	Convolutional Neural Network
DL	Deep Learning
FPS	Frames Per Second
GaaS	Government as a Service
GPU	Graphics Processing Unit
ID	Identifier
IoT	Internet of Things
IoU	Intersection over Union
ITS	Intelligent Transportation System
ITMS	Intelligent Traffic Management System
JSON	JavaScript Object Notation
LED	Light-Emitting Diode
mAP	Mean Average Precision
ML	Machine Learning
MOTA	Multi-Object Tracking Accuracy
NMS	Non-Maximum Suppression
REST	Representational State Transfer
SORT	Simple Online and Realtime Tracking
TVDS	Traffic Violation Detection System
WHO	World Health Organization
YOLO	You Only Look Once

1. INTRODUCTION

Road traffic violations are among the most preventable causes of accidents, injuries, and deaths on the world's roads. According to the World Health Organization, road crashes claim approximately 1.19 million lives every year and leave tens of millions more with serious injuries [1]. In Sri Lanka, the situation is acute. Despite government investments in road safety campaigns and infrastructure improvements, the country recorded 2,813 road fatalities and over 24,500 injuries in 2022 alone placing it among the highest road fatality rates per capita in South Asia [2]. Behind these statistics lie patterns that point directly to the violations this system is designed to address: red light running at signalised intersections and dangerous lane weaving on multi-lane roads. Research by the Sri Lanka Police Traffic Division indicates that around 22% of serious crashes are intersection-related many involving red-light violations while 3333wwwslane indiscipline including weaving contributes to a further 17% [3]. Together, these two violation types are implicated in nearly two-fifths of all serious road accidents in Sri Lanka.

The enforcement gap is equally stark. With a vehicle fleet that has grown at an average annual rate of 7.3% over the past decade reaching over 8.5 million registered vehicles by 2023 the demands on traffic enforcement have increased far faster than the resources available to meet them [2]. Manual police enforcement, while valuable, is fundamentally limited by human presence: an officer at one location cannot simultaneously monitor other roads. This creates vast blind spots where violations occur unchecked, with no evidence, no record, and no consequence for the offending driver. The result is not just an enforcement failure it is a signal to every driver on the road that the rules may be safely ignored if no officer happens to be present.

Automated traffic violation detection using computer vision and artificial intelligence offers a credible solution to this problem. Modern deep learning systems can analyse live video from roadside cameras, detect every vehicle in the frame in milliseconds, track those vehicles consistently across frames, and classify whether a traffic violation has occurred all in real time, all without human involvement. When a violation is confirmed, the system can immediately record the event, link it to the responsible

driver through the vehicle's registration number, update the driver's safety score, and deliver a notification to their mobile device creating a complete enforcement and feedback loop in a matter of seconds.

This report presents the Traffic Violation Detection and Driver Safety Scoring System (TVDS), developed as one component of the Intelligent Traffic Management System (ITMS) built by Group 25-26J-330 at the Sri Lanka Institute of Information Technology. The TVDS uses computer vision techniques to analyse live traffic video streams and detect violations in real time. Once a violation is detected, it is recorded and linked to the corresponding driver, and the driver's safety score is updated accordingly. This creates a transparent and fair mechanism to monitor driving behaviour over time, and to encourage safer habits through continuous, personalised feedback.

1.1 Background and Motivation

Understanding why this system is needed requires looking at the full picture of traffic enforcement in Sri Lanka. The country's vehicle fleet has grown dramatically over the past decade. Between 2015 and 2023, the total number of registered vehicles increased from under 5 million to over 8.5 million, with motorcycles alone accounting for 3.8 million registrations [2]. Road infrastructure has not expanded at a comparable pace, and enforcement resources have struggled even more to keep up. The resulting environment dense, fast-moving, mixed-vehicle traffic on roads not designed for such volumes is one where violations are common and detection is sporadic.

The consequences are severe and well-documented. Intersection-related accidents, many involving red light violations, consistently rank among the leading causes of fatalities in the Western Province, where Colombo's urban traffic is most concentrated. Highway incidents linked to lane indiscipline including weaving, tailgating, and sudden unsingable lane changes account for a disproportionate share of high-speed fatal crashes on the Southern Expressway and other modern road links [3]. These are not problems that awareness campaigns alone can solve. They require a deterrent that operates consistently, on a scale, and at the point where dangerous behavior occurs.

Conventional CCTV systems, while widely deployed, do not meet this standard. They record footage that is reviewed after the fact if it is reviewed at all and their technical capabilities for automated detection are limited to simple rule-based analytics that cannot handle the diversity and complexity of real Sri Lankan traffic. What is needed is a system that can learn to recognise vehicles and their behaviours from data, operate reliably under variable conditions, and translate every detected violation into an immediate, fair consequence. That is precisely the goal this component was designed to achieve.

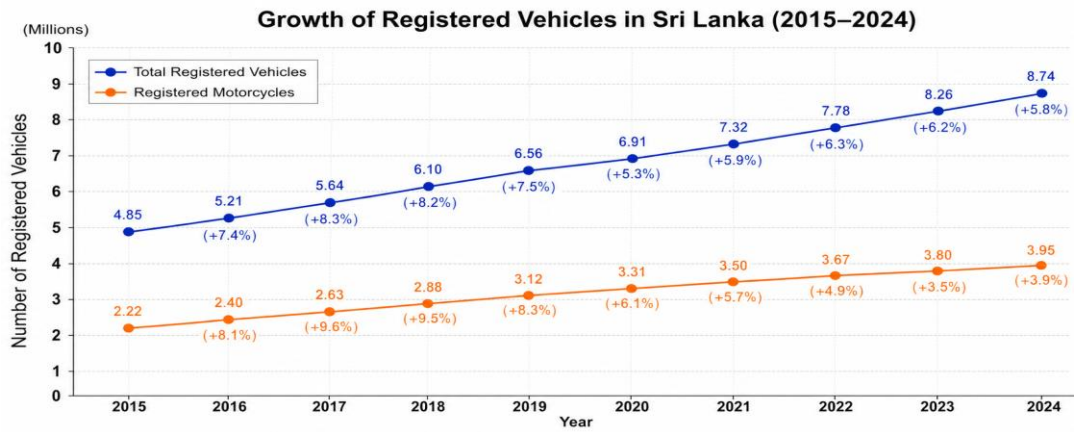


Figure 1.1: Growth of registered vehicles in Sri Lanka 2015–2024

1.2 Literature Survey

1.2.1 Traditional traffic monitoring and its limitations

The history of automated traffic monitoring stretches back several decades. Early systems relied on inductive loop detectors embedded in road surfaces wire loops that detect changes in inductance as a vehicle passes over them. These systems could count vehicles and estimate speed, but were blind to lateral behaviour, could not identify vehicle types, and required expensive and disruptive civil engineering works to install and maintain. As a result, their use was largely limited to high-priority locations such as major junctions and expressway entry points [4].

The introduction of camera-based surveillance systems in the 1990s and 2000s expanded monitoring capability significantly. Cameras could observe larger areas, capture visual evidence of violations, and in theory support automated analysis through computer vision techniques. In practice, however, the computer vision methods available at the time proved brittle. Systems based on background subtraction, which model the expected background appearance of a scene and flag moving objects as foreground, failed when lighting changed across the day or when the camera view contained moving elements like tree branches or flags. Optical flow estimation, which tracks the apparent motion of pixel patterns between frames, was computationally expensive and sensitive to noise. Blob-based vehicle detection, which identified connected regions of changed pixels as vehicle candidates, worked adequately for simple counting but could not reliably distinguish vehicle types, handle overlapping vehicles, or detect the behavioural patterns associated with specific violations [5].

The net result was that camera-based automated enforcement remained limited in deployment and reliability. Most systems could detect only simple, unambiguous violations a vehicle crossing a stop line during a clear red signal, for instance but could not handle more nuanced violations like sustained lane weaving, near-miss scenarios, or violations involving partially occluded vehicles. The enforcement gap remained open.

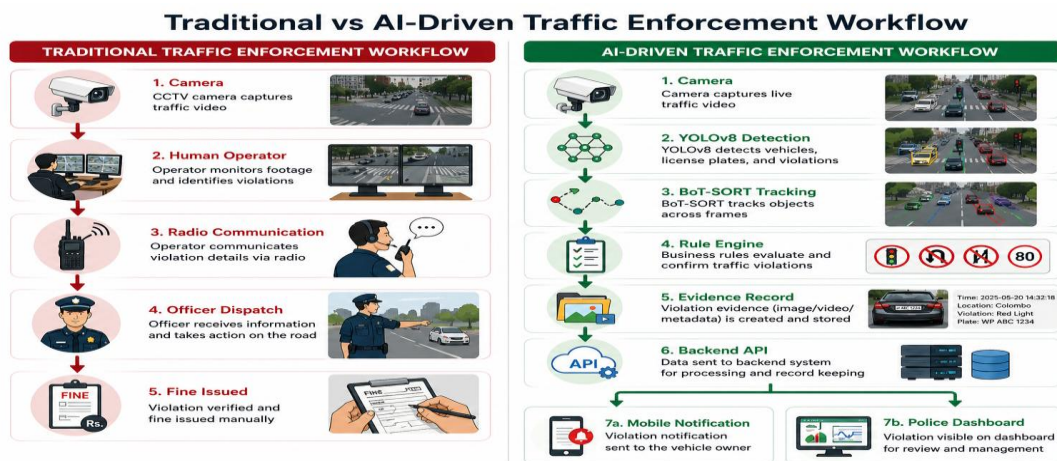


Figure 1.2: Traditional vs AI-driven traffic enforcement workflow

1.2.2 Deep learning for vehicle detection

The emergence of deep convolutional neural networks transformed automated visual analysis. Where classical computer vision required engineers to manually specify features edges, corners, textures that the system should look for, deep learning networks discover these features automatically during training. Given enough labelled examples, a deep network learns to represent visual patterns at multiple levels of abstraction: simple edges at the earliest layers, complex object parts in intermediate layers, and whole objects at the final layers. This end-to-end learning approach proved dramatically more robust to the kind of variability changing lighting, weather, viewpoints that defeated classical methods [6].

Early deep detection systems like R-CNN and its successors used a two-stage approach: first generating a set of candidate regions that might contain objects, then classifying each candidate. These systems achieved high accuracy but were too slow for real-time applications, often taking several seconds per frame. The development of single-stage detectors networks that predict bounding boxes and class labels in a single forward pass brought detection speeds into the range needed for live video analysis [7].

Of these single-stage architectures, the You Only Look Once (YOLO) family has become the dominant approach for real-time traffic surveillance. Its combination of speed and accuracy, along with an active open-source development community, has made it the practical standard for systems requiring real-time, multi-class object detection from video streams. The latest iteration, YOLOv8, introduced significant architectural improvements including an anchor-free detection head and decoupled classification and regression branches, achieving state-of-the-art performance on standard benchmarks while maintaining real-time throughput on edge hardware [8].

1.2.3 multi-object tracking

Detecting vehicles in individual video frames is only the first step toward violation analysis. Understanding whether a specific vehicle has run a red light or weaved through traffic requires following that vehicle consistently across many consecutive frames the multi-object tracking problem. This is technically challenging: vehicles

occlude one another, temporarily leave and re-enter the camera field of view, and can appear visually similar. Any of these events can cause a tracking system to lose or confuse vehicle identities, corrupting the trajectory history needed for violation classification [9].

The foundational tracking approach, SORT (Simple Online and Realtime Tracking), combines a Kalman filter which predicts where each tracked vehicle should be in the next frame based on its current velocity with the Hungarian algorithm for matching new detections to predicted positions. SORT is fast and lightweight but can lose track identities under occlusion because it matches vehicles based only on position, with no memory of what they look like [10].

BoT-SORT (Bootstrapped Object Tracking with SORT), the algorithm selected for this component, addresses SORT's limitations with two important additions. A camera motion compensation module detects and corrects for apparent displacement of tracked objects caused by camera vibration or wind sways a practical issue for pole-mounted roadside cameras. A visual re-identification embedding supplements position-based matching with appearance similarity, enabling the tracker to correctly re-identify vehicles after brief occlusion events. These additions substantially reduce the identity switch rate that degrades classical SORT performance in dense, real-world traffic [11]. ByteTrack contributed the idea of using low-confidence detections in a second matching round to recover partially occluded vehicles [12], and this approach is incorporated into BoT-SORT's matching pipeline.

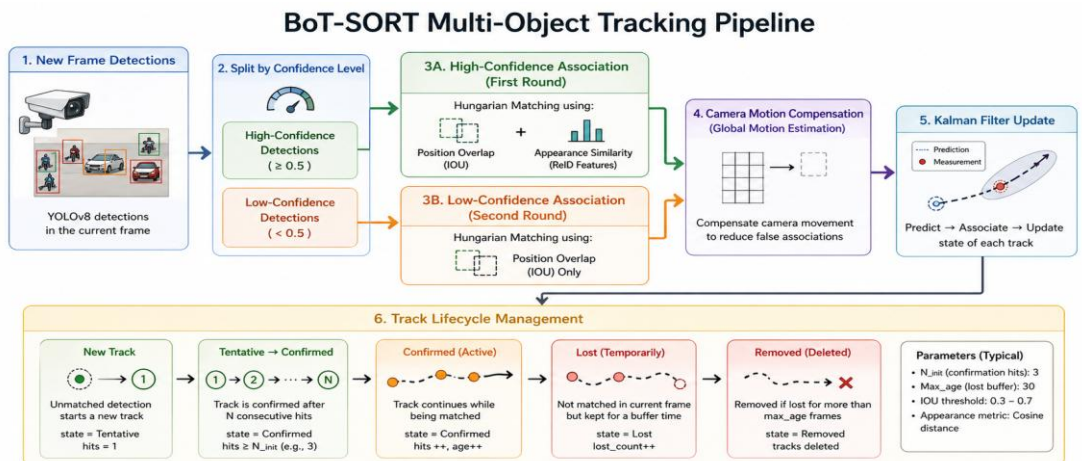


Figure 1.3: BoT-SORT multi-object tracking pipeline

1.2.4 Traffic violation detection systems

Published research on automated traffic violation detection has grown considerably in recent years, driven by improvements in detection accuracy and the falling cost of edge computing hardware. Chen et al. demonstrated red light violation detection using a deep learning detector combined with optical flow tracking, achieving 85% accuracy on Chinese intersection footage, though the system's 120-millisecond per-frame inference time made real-time deployment impractical [13]. Singh and Gupta applied YOLOv5 to Indian intersection monitoring and reached 89.2% detection accuracy, but evaluation was limited to clear daytime conditions [14]. Zhang et al. demonstrated YOLOv8's advantage for small-vehicle detection particularly motorcycles achieving 93.1% mAP on a Chinese multi-class vehicle dataset [15]. Luvizon et al. addressed lane change detection from roadside cameras, but their approach did not distinguish persistent oscillatory weaving from a deliberate single lane change [16].

A common limitation across this body of work is its focus on individual violation types in isolation. No published system combines red light and lane weaving detection in a single pipeline adapted for Sri Lankan Road conditions, and no published system connects violation detection to a real-time personalized driver feedback mechanism. These gaps define the specific contribution of this component.

1.2.5 Driver feedback and behavioral scoring

Research in transportation psychology consistently demonstrates that the timing and format of feedback strongly influences its effectiveness in modifying behavior. Delayed administrative penalties fines received days or weeks after a violation are significantly less effective at changing behavior than immediate, personalized feedback delivered at or near the time of the infraction [17]. The psychological connection between behavior and its consequence weakens rapidly with time, particularly when the consequence is financial rather than experiential.

Gamified scoring systems have emerged as one of the most promising approaches to sustained behavioural feedback in road safety applications. Usage-based insurance programmes and eco-driving applications that present fuel efficiency as a real-time score consistently produce measurable improvements in driver behaviour compared to

baseline groups [18]. The Driver Safety Score in this component draws directly on this evidence: a 100-point daily score that changes immediately when a violation occurs, communicated through a mobile application the driver checks regularly, creates a feedback loop that is both immediate and personally meaningful. The daily reset mechanism ensures the score reflects recent behaviour rather than permanent history, giving drivers a fresh opportunity to improve each day.

1.3 Research Gap

Five clear gaps in existing research and practice motivated the development of this component.

The first gap is the absence of any deployed system in Sri Lanka that combines red light and lane weaving violation detection in a single unified pipeline. Handling both violation types through one system on a single edge device is more efficient, more consistent, and more cost-effective than deploying separate specialist systems at each location.

The second gap is the reliance of most published systems on generic training datasets that do not include the vehicle types of characteristics of Sri Lankan roads particularly three-wheelers (tuk-tuks) and the diverse range of motorcycle models common in the country. A model trained on Chinese or European traffic data will produce higher error rates when deployed in Sri Lanka than one trained on locally representative data.

The third gap is the absence of real-time personalized driver feedback in existing enforcement systems. Current Sri Lankan enforcement creates consequences fines, license points but delivers them days or weeks after the violation through administrative channels. The delay reduces their effectiveness as behavior change tools.

The fourth gap is the absence of a complete, end-to-end system that spans from violation detection all the way through to driver notification, fine payment, and long-term score tracking in a single integrated workflow. Most published systems stop at the detection step.

The fifth gap is the lack of IoT-based immediate on-site alerting in published Sri Lankan enforcement technology. Instant physical alerts at the point of violation LED indicators and auditory buzzers complement digital notifications to create a multi-channel feedback mechanism that reaches drivers both in the moment and through their mobile device.

Research Gap	Existing Systems	This Component
Combined red light + weaving detection	Single violation type per system	Both types unified in one pipeline
Sri Lankan vehicle class coverage	Generic global datasets	Locally annotated SL training data
Real-time personalised driver feedback	No SL system identified	Live score update + push notification
End-to-end enforcement workflow	Detection output only	Detection → record → score → payment
IoT on-site alerts	Absent from most systems	LED and buzzer at violation point

Table 1.1: Research gap analysis

1.4 Research Problem

The research problem this component addresses can be stated as follows: how can a camera-based AI system reliably detect red light running and dangerous lane weaving violations in real time, under the variable conditions of Sri Lankan urban traffic, and immediately translate every confirmed violation into a personalized, transparent consequence for the driver one that is proportionate, fair, and delivered quickly enough to influence future behavior?

This problem has four specific technical dimensions. The detection model must accurately identify all vehicle types common on Sri Lankan roads, including three-wheelers and motorcycles that are absent from standard training datasets. The tracking system must maintain consistent vehicle identities through the occlusion and high-density characteristic of peak-hour urban traffic. The violation rule engine must classify violations accurately enough to avoid both false positives which are unjust and undermine trust and false negatives which undermine the deterrent effect. And the complete pipeline must process video at 30 frames per second on resource-constrained edge hardware deployed at roadside locations without reliable high-bandwidth internet connectivity.

1.5 Research Objectives

1.5.1 Main objective

The main objective of this component is to design, implement, and evaluate a real-time Traffic Violation Detection and Driver Safety Scoring System that uses YOLOv8 for vehicle detection and BoT-SORT for multi-object tracking to accurately identify red light running and lane weaving violations from live traffic camera feeds, and delivers violation records, safety score deductions, and driver notifications to the ITMS backend and companion mobile application in real time.

1.5.2 Specific objectives

1. Train and fine-tune a YOLOv8 model on a locally collected Sri Lankan traffic dataset covering seven vehicle classes, targeting mAP@0.5 of at least 88%.
2. Integrate BoT-SORT multi-object tracking with camera motion compensation to assign stable unique IDs to vehicles and maintain trajectory records, targeting an ID switch rate below 5% on test sequences with more than 20 simultaneous tracks.
3. Develop a violation rule engine that classifies red light running with precision and recall both above 90%, and lane weaving with precision and recall both above 85%, under normal daytime conditions.

4. Implement a gamified 100-point daily driver safety score that deducts 15 points per red light violation and 5 points per lane weaving event, with automatic nightly reset.
5. Design and deploy an eight-screen driver mobile application delivering real-time violation notifications, safety score display, violation history, and online fine payment.
6. Integrate IoT-based LED and buzzer alerts at violation points to provide immediate on-site feedback.
7. Demonstrate end-to-end pipeline processing at 30 frames per second on target edge computing hardware.

2. METHODOLOGY

This chapter describes the complete design and implementation of the Traffic Violation Detection and Driver Safety Scoring System. Each subsection covers one functional element of the component, explaining what it does, how it works, and why the specific design choices were made.

2.1 Component Overview

This component focuses on automatically detecting traffic violations and evaluating driver behaviour using an intelligent scoring system. The goal is not only to identify violations in real time but also to encourage safer driving habits through continuous monitoring and feedback. The system uses computer vision techniques to analyse live traffic video streams and detect violations such as red light running and lane weaving. Once a violation is detected, it is recorded and linked to the corresponding driver, and the driver's safety score is updated accordingly. This creates a transparent and fair mechanism to monitor driving behaviour over time.

The component sits within the broader Intelligent Traffic Management System developed by Group 25-26J-330, alongside modules for adaptive signal control, parking violation detection, and accident prediction. The TVDS is the component responsible for moving violations infractions committed by vehicles while in motion.

It receives live video from roadside cameras, processes it through the detection and tracking pipeline, evaluates trajectories for violation patterns, and communicates confirmed violations to the shared ITMS backend and to drivers through the mobile application.

The design philosophy of this component rests on three principles. First, detection must be reliable enough to withstand legal scrutiny false positives are not merely inconvenient, they are unjust. Second, the feedback loop must be fast enough to influence behavior of a notification received seconds after a violation is more effective than a fine letter received weeks later. Third, the system must be practical enough to deploy and maintain at scale complex infrastructure that requires constant specialist attention will not survive real-world deployment conditions.

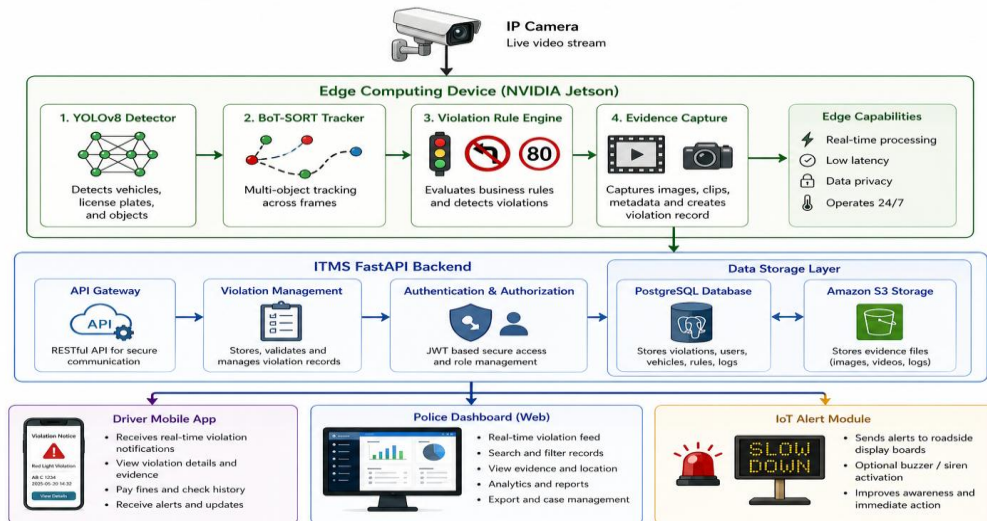


Figure 2.1: High-level system architecture of the TVDS component'

2.2 System Architecture

2.2.1 High-level architecture

The TVDS is structured as a processing pipeline running on an edge computing device co-located with the roadside camera. This edge-first design was chosen for two reasons: it minimises the amount of raw video data that must be transmitted over the network (only structured violation records and compressed evidence images are sent, not continuous video streams), and it allows the system to continue functioning even when network connectivity is degraded or temporarily unavailable.

The pipeline begins with video capture from an IP camera via an RTSP stream. Frames are processed by YOLOv8, which identifies all vehicles present and outputs their bounding boxes, class labels, and confidence scores. BoT-SORT then assigns or updates a unique integer track ID for each detected vehicle and updates the trajectory record for each active track. The violation rule engine evaluates each vehicle's current position and trajectory history to determine whether a violation condition has been met. When a violation is confirmed, the evidence capture module assembles a record containing video frames and metadata, which is transmitted to the central ITMS FastAPI backend. The backend stores the record, updates the driver's safety score, and triggers a push notification to the driver's mobile device. Simultaneously, the IoT alert module triggers the LED and buzzer at the violation site.

2.2.2 Data flow

The data flow through the system follows a clear sequence. The IP camera streams video at 30 frames per second to the edge device. Each frame enters the detection pipeline, where YOLOv8 processes it and outputs a list of detected vehicle bounding boxes with class labels and confidence scores. These detections are passed to BoT-SORT, which matches them to existing tracks, updates trajectory records, and creates new tracks for newly appeared vehicles. The trajectory data is then evaluated by the violation rule engine, which checks each vehicle's position against defined violation zones and examines trajectory history for patterns consistent with lane weaving. When a violation is confirmed, a structured JSON violation record is prepared and queued for transmission. The backend receives this record, performs driver account lookup by

licence plate number, applies the score deduction, and dispatches a push notification. In parallel, the IoT controller sends the signal to activate the LED indicator and buzzer at the roadside installation.

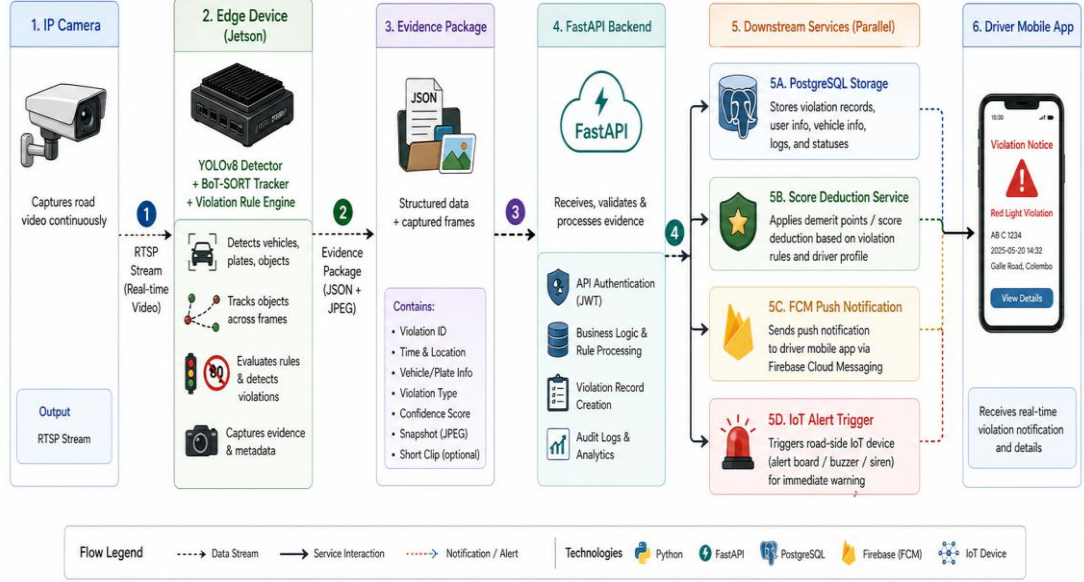


Figure 2.2: System data flow camera feed to driver notification

2.3 Data Collection and Dataset Preparation

The quality of a deep learning detection model depends directly on the quality and representativeness of the data it is trained on. A model trained only on Chinese or European traffic footage will perform poorly in Sri Lanka because the vehicle types, road markings, lighting conditions, and traffic density patterns are different. For this reason, data collection was treated as a foundational priority rather than an afterthought.

Training data was assembled from two sources. A base layer of publicly available benchmark datasets including UA-DETRAC [19] and a filtered subset of BDD100K [20] provided a diverse range of vehicle appearances across different countries and lighting conditions. This global base was supplemented with approximately 42 hours of locally recorded footage from three signalised intersections in the Colombo

Metropolitan Area and two segments of the Southern Expressway. Local footage was recorded across multiple time windows covering morning peak, midday, evening peak, and late night, and under weather conditions spanning clear, overcast, light rain, and heavy rain.

Camera Location	Type	Recording Duration	Conditions Captured
Baseline Road Junction, Colombo	Intersection	14 hours	Clear, overcast, rain, night
Kirillapone Junction, Colombo	Intersection	12 hours	Clear, overcast, rain
Peliyagoda Flyover Approach	Intersection	10 hours	Peak hour, off-peak, night

Table 2.1: Camera hardware specifications

From the combined footage, 28,500 frames were extracted at regular intervals and annotated using LabelImg in YOLO format. Seven vehicle classes were labelled: motorcycle, three-wheeler (tuk-tuk), car, van, bus, truck, and other. The inclusion of three-wheelers as a distinct class was a deliberate design decision this class is entirely absent from standard benchmark datasets yet constitutes a major share of Sri Lankan urban traffic. Two annotators worked independently on a 2,000-frame validation subset, and their agreement (Cohen's kappa = 0.87) confirmed that labelling quality was sufficient for reliable model training.

Vehicle Class	Annotated Instances	Percentage of Total	Difficulty
Motorcycle	8,240	28.9%	High small, fast, frequent occlusion
Three-wheeler	5,180	18.2%	Medium unique shape, locally common

Car	7,620	26.7%	Low large, visually consistent
Van	2,890	10.1%	Low large, distinct silhouette
Bus	1,740	6.1%	Low very large, easy to identify
Truck	1,320	4.6%	Low large, distinctive
Other	1,510	5.3%	Medium varied appearance

Table 2.2: Dataset annotation summary by vehicle class

Data augmentation during training increased the effective diversity of the training set without requiring additional data collection. Augmentation techniques applied included horizontal flip at 50% probability, mosaic augmentation combining four training images into a single composite at 100% probability for the first 120 training epochs, HSV colour jitter to simulate variable lighting and weather conditions, random scale changes of plus or minus 50% to handle varying vehicle distances, and MixUp blending at 10% probability. These augmentations ensured the model encountered the kind of variability occlusion, glare, rain, mixed densities that characterizes real deployment conditions.

2.4 Vehicle Detection Using YOLOv8

2.4.1 Why YOLOv8

YOLOv8 is used for real-time vehicle detection due to its high accuracy and speed. The model identifies different types of vehicles and detects violations directly from live video streams. Of all the detection architectures evaluated during the design phase, YOLOv8 offered the best combination of detection accuracy, inference speed on edge hardware, and availability of an actively maintained open-source implementation [8].

YOLOv8 processes an entire video frame in a single forward pass through the network, predicting bounding boxes and class probabilities simultaneously without requiring a separate region proposal stage. This single-pass architecture is what enables YOLOv8

to operate at the frame rates needed for real-time surveillance. The architecture consists of three main parts: a CSP (Cross-Stage Partial) backbone built from C2f bottleneck modules that extracts hierarchical visual features from the input frame; a PANet (Path Aggregation Network) neck that aggregates features across different scales to enable detection of both small vehicles (motorcycles at distance) and large vehicles (buses in close proximity); and an anchor-free detection head that directly predicts bounding box coordinates and class probabilities without requiring pre-defined anchor boxes, which reduces sensitivity to the choice of anchor shapes and improves performance on unusual vehicle aspect ratios.

The anchor-free design is particularly important for Sri Lankan traffic conditions. Three-wheelers have aspect ratios that differ substantially from standard cars and vans, and historical anchor-based detectors often struggled with these unconventional shapes. By removing the dependency on predefined anchors, YOLOv8 handles unusual aspect ratios more naturally, contributing to the improved three-wheeler detection accuracy observed in evaluation.

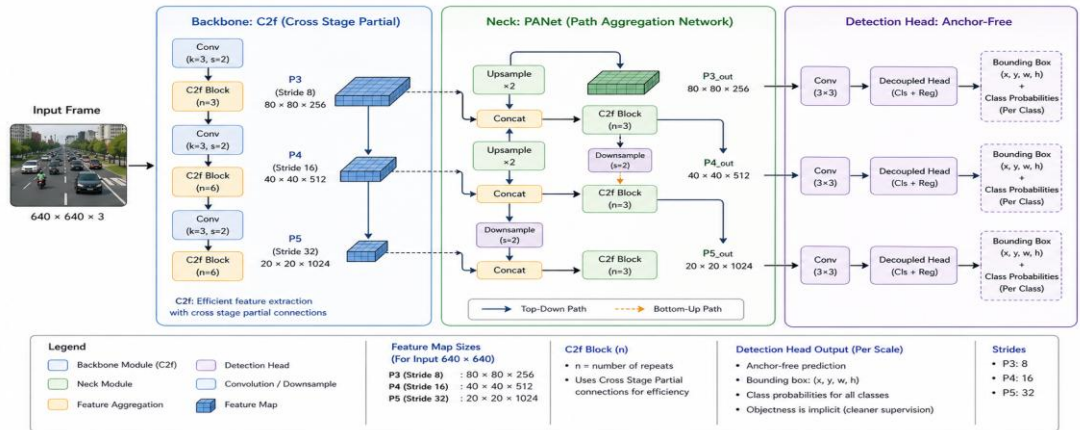


Figure 2.4: YOLOv8 architecture backbone, neck, and detection head

2.4.2 Model training and configuration

YOLOv8 is available in five scale variants nano (n), small (s), medium (m), large (l), and extra-large (x) offering different trade-offs between detection accuracy and computational cost. All five variants were benchmarked on the target deployment hardware to identify which met the 30 FPS throughput requirement while maximizing

Variant	Parameters (M)	mAP@0.5 (COCO)	FPS on Target Hardware	Decision
YOLOv8n	3.2	37.3%	148 FPS	No accuracy insufficient
YOLOv8s	11.2	44.9%	98 FPS	No borderline accuracy
YOLOv8m	25.9	50.2%	61 FPS	Selected best balance
YOLOv8l	43.7	52.9%	38 FPS	No throughput insufficient
YOLOv8x	68.2	53.9%	24 FPS	No throughput insufficient

detection accuracy.

Table 2.3: YOLOv8 model variant comparison on target hardware

YOLOv8m was selected as offering the best balance between detection accuracy and real-time throughput. It was fine-tuned for 150 epochs on the combined dataset using stochastic gradient descent with momentum, a cosine annealing learning rate schedule beginning at 0.01 and reducing to 0.001 over the training period. After fine-tuning on local Sri Lankan data, the model achieved mAP@0.5 of 91.4% on the held-out test set a meaningful improvement over the COCO-pretrained baseline applied without fine-tuning, confirming the value of domain-specific training data.

2.4.3 Detection output and post-processing

The raw output of YOLOv8 for each frame is a set of candidate bounding box predictions at three detection scales, corresponding to small, medium, and large vehicles. These raw predictions are post-processed using Non-Maximum Suppression (NMS) to remove redundant overlapping boxes: where multiple boxes predict the same physical vehicle, only the highest-confidence box is retained. Boxes below a minimum confidence threshold are discarded as likely false positives. The resulting clean set of bounding boxes each with a class label and confidence score is passed to the tracking module.

2.5 Vehicle Tracking Using BoT-SORT

2.5.1 Tracking algorithm overview Each vehicle is assigned a unique ID through BoT-SORT, which helps avoid duplicate detection and ensures accurate violation tracking. BoT-SORT maintains a pool of active track objects, one per physical vehicle currently in the camera field of view. Each track stores a unique integer ID, a Kalman filter state vector encoding the vehicle's estimated position, size, and velocity, a history buffer of recent trajectory positions, a visual appearance embedding for re-identification, and a lifecycle counter.

At each frame, tracking proceeds in three steps. First, the Kalman filter for each active track predicts where that vehicle should be in the current frame, based on its position and velocity from previous frames. Second, the Hungarian assignment algorithm matches new YOLOv8 detections to predicted track positions, using a cost matrix that considers both the geometric overlap between predicted and detected bounding boxes and the similarity between the track's appearance embedding and the appearance of the new detection. High-confidence detections are matched first. Remaining low-confidence detections which would normally be discarded as likely false positives are offered a second matching round using position overlap only, following the ByteTrack approach [12]. This two-round matching recovers vehicles that are partially occluded and thus detected with lower confidence, significantly improving tracking continuity in dense traffic. Third, tracks that remain unmatched for more than 30 consecutive frames are retired and their IDs freed for reuse.

Parameter	Value	Purpose
High-confidence threshold	0.5	Minimum confidence for first-round matching
Low-confidence threshold	0.1	Minimum confidence for second-round matching
Position overlap threshold (high conf.)	0.8	Tight association in dense traffic
Position overlap threshold (low conf.)	0.5	Looser association for partial occlusion
Maximum frames before track retirement	30 (1 second at 30 FPS)	Retire genuinely lost tracks
Minimum detections to confirm track	3 consecutive frames	Filter single-frame noise
Visual embedding dimension	128 dimensions	Compact for edge hardware
Trajectory history buffer length	90 frames (3 seconds)	Sufficient for weaving analysis

Table 2.4: BoT-SORT configuration parameters

2.5.2 Unique ID assignment and trajectory management

Each vehicle that enters the camera field of view and remains detectable for at least three consecutive frames is assigned a unique integer track ID. This ID is maintained for the entire time in the monitored zone, even if it is briefly occluded. The assignment of a stable, unique ID is what enables the violation rule engine to analyse the trajectory history of a specific vehicle without consistent IDs, the engine cannot distinguish the movement of one vehicle from another, and violations involving occlusion or brief tracking loss would be missed or misattributed.

Each track maintains a trajectory buffer, a record of the vehicle's position in each frame over the last 90 frames (3 seconds at 30 FPS). This 3-second window was chosen after analysis of real weaving events from the collected footage: the vast majority of genuine weaving episodes complete a detectable oscillation pattern within 2.5 seconds, meaning a 3-second window captures sufficient history with some safety margin. For red light violation detection, only the current position relative to the defined intersection zones matters trajectory history is not needed so the rule engine accesses only the current frame's position data for that analysis.

2.5.3 Camera motion compensation

Camera motion compensation (CMC) is one of BoT-SORT's most practically important features for infrastructure-mounted deployment. Traffic signal poles, gantry structures, and the wall mounts used for roadside cameras are subject to vibration caused by heavy vehicles passing, wind loading, and minor structural movement. Even small apparent displacements of all tracked objects between frames caused by the camera moving, not the vehicles can corrupt Kalman filter predictions and cause the assignment algorithm to make errors.

The CMC module estimates global background motion between consecutive frames by computing sparse optical flow on stable regions of the image sky, buildings, road infrastructure that should not move between frames. The estimated displacement is subtracted from each track's Kalman prediction before matching, effectively removing camera-induced drift. In testing at sites with detectable camera vibration, disabling CMC increased identity switches by more than three times, confirming that this feature is not optional for reliable performance at real-world deployment locations.

2.6 Violation Detection Rule Engine

The violation detection rule engine analyses the trajectory and position data produced by the detection and tracking pipeline to determine whether a traffic violation has occurred. The engine handles two violation types: red light running and lane weaving. Each type has its own detection logic, but both share the same underlying principle a violation is confirmed only when a specific set of spatial and temporal conditions have been met, designed to be specific enough to avoid false positives

while being sensitive enough to catch genuine violations.

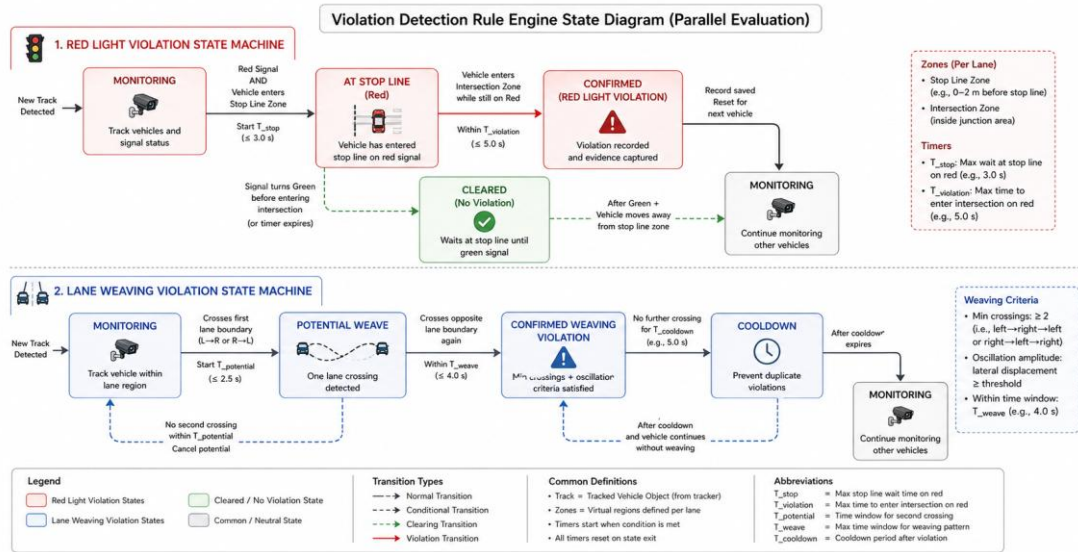


Figure 2.5: Violation detection rule engine state diagram

2.6.1 Red light violation detection

Red light violation detection works by cross-referencing a vehicle's position with two spatial zones defined at each monitored intersection, together with the current state of the traffic signal controlling that approach.

Two zones are defined for each monitored intersection approach. The stop-line zone is a spatial region around the physical stop line of the intersection. A vehicle whose position enters this zone while the signal is red is flagged as a potential violator. The intersection zone covers the full intersection area beyond the stop line. A vehicle that enters the intersection zone after having been flagged as a potential violator is confirmed as a genuine red-light violator.

This two-zone approach is important for system fairness. A vehicle that enters the stop-line zone on a red signal but stops before crossing into the intersection perhaps because the driver braked hard at the last moment does not generate a confirmed violation. Only vehicles that actually enter the intersection area during a red signal phase are confirmed as violators. This distinction is critical: flagging a vehicle that stopped at the line as a red-light runner would be unjust and would immediately undermine public trust in the system.

The traffic signal state is obtained through integration with the intersection's signal controller system, which exposes the current phase and remaining duration via an API endpoint. This is provided by the adaptive signal controller component developed by a teammate within the ITMS. Where direct controller integration is unavailable, a camera-based signal state detector, a classifier trained on images of the signal head showing red, amber, and green states provides a fallback. A short time alignment window accounts for any delay between signal state changes and their detection by the pipeline.

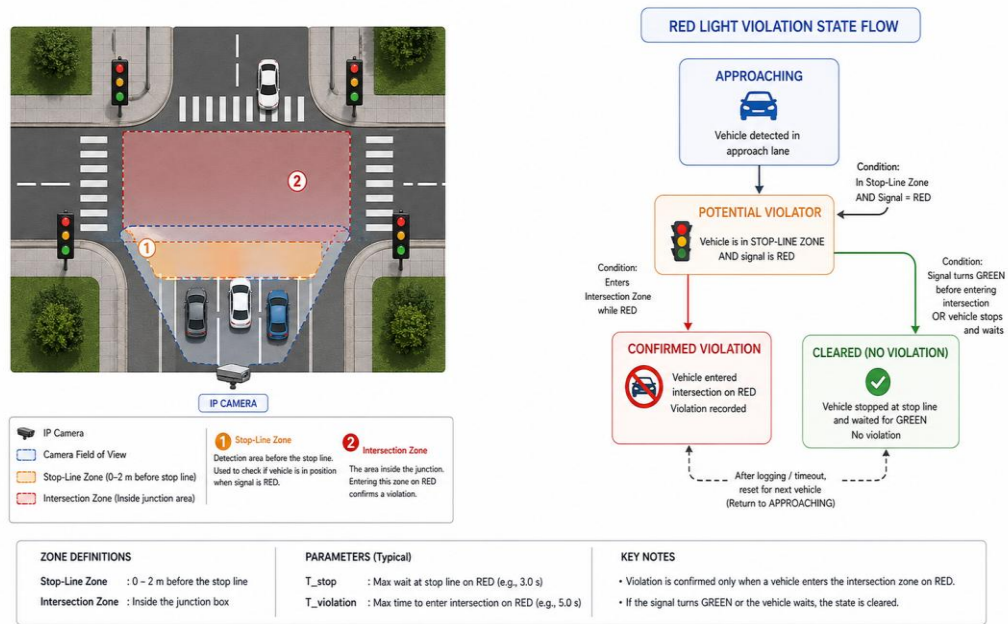


Figure 2.6: Red light violation detection zone and state flow

2.6.2 Lane weaving detection

Lane weaving detection analyses the trajectory history of each tracked vehicle to identify the oscillatory lateral movement pattern that defines dangerous weaving. The distinction between a legal lane change which involves crossing from one lane to an adjacent lane in a controlled, intentional manner and illegal lane weaving which involves repeated, erratic lateral movement across lane boundaries is one of the most technically challenging aspects of violation detection.

The rule engine evaluates each vehicle's trajectory buffer at every frame. Lane boundaries for the monitored road segment are defined as a set of lines in the image

plane, corresponding to the visible solid lane divider markings. The engine tracks when the vehicle's position crosses these boundaries and in which direction. A lane weaving violation flag is raised when three conditions are all satisfied: the number of boundary crossings in the trajectory window exceeds a minimum threshold set higher than what a single deliberate lane change produces; the crossings are oscillatory there are crossings in both the left and the right directions, ruling out a smooth unidirectional lane change; and the vehicle's speed is above a minimum threshold, excluding stopped or very slow vehicles whose lateral drift is unlikely to represent genuine weaving. A cooldown timer prevents the same weaving episode from generating multiple violation records.

Parameter	Value	Justification
Minimum crossing events	3 crossings	Legal lane change produces 1 crossing only
Required crossing directions	Both left and right	Rules out unidirectional lane changes
Trajectory evaluation window	90 frames (3 seconds)	Sufficient for one complete weave cycle
Post-violation cooldown timer	10 seconds	Prevents duplicate records per episode
Minimum vehicle speed	15 km/h	Excludes very slow or stationary vehicles

Table 2.5: Violation detection rule parameters

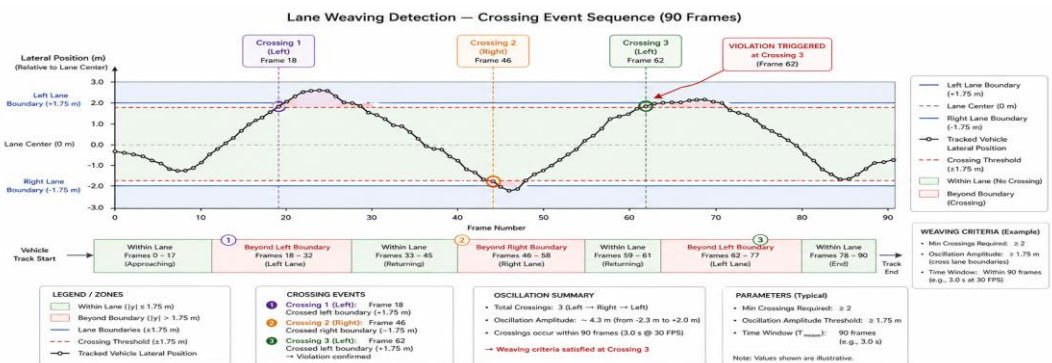


Figure 2.7: Lane weaving detection crossing event sequence

2.6.3 Preventing duplicate detection

A key requirement of the violation detection system is that each distinct violation event is recorded only once. Without explicit de-duplication logic, a single weaving episode lasting several seconds could generate dozens of redundant records as the rule engine evaluates the same trajectory window repeatedly across successive frames. Similarly, a vehicle that remains in the intersection zone for several seconds after crossing the stop line on red could generate multiple records for the same single infraction.

De-duplication is handled at two levels. At the track level, each vehicle's track maintains a per-violation-type timestamp recording when the last violation was confirmed. A new violation of the same type for the same track is not raised until the cooldown period has elapsed. At the system level, violation records include the track ID and violation timestamp, and the backend API performs a duplicate check before storing any new record if a record with the same track ID, violation type, and a timestamp within 30 seconds of an existing record is received, it is flagged as a potential duplicate and reviewed rather than stored directly.

2.7 Evidence Capture and Recording

When a violation is confirmed by the rule engine, the evidence capture module assembles a structured record of the event. This record serves two purposes: it provides the factual basis for the administrative enforcement action (fine, score deduction), and it provides the driver with transparent, verifiable evidence of what the system recorded.

The evidence package contains two categories of data. The first is video evidence: a sequence of frames from the moment of violation, specifically five frames immediately before the violation event, the violation frame itself (the frame in which the rule engine raised the confirmed flag), and five frames immediately after. These eleven frames are compressed as JPEG images and included in the violation record. The selection of frames before and after the violation event provides context showing the vehicle's approach, the violation, and its immediate aftermath which is important both for administrative review and for driver understanding.

The second category is metadata: a structured JSON record containing the violation type, timestamp in UTC format, camera identifier, GPS coordinates of the camera location, the BoT-SORT track ID of the violating vehicle, the YOLOv8-classified vehicle class, the ANPR-extracted licence plate number (when readable), and the score deduction amount. This record is transmitted to the FastAPI backend via an HTTP POST request and acknowledged with a 201 Created response when successfully stored.

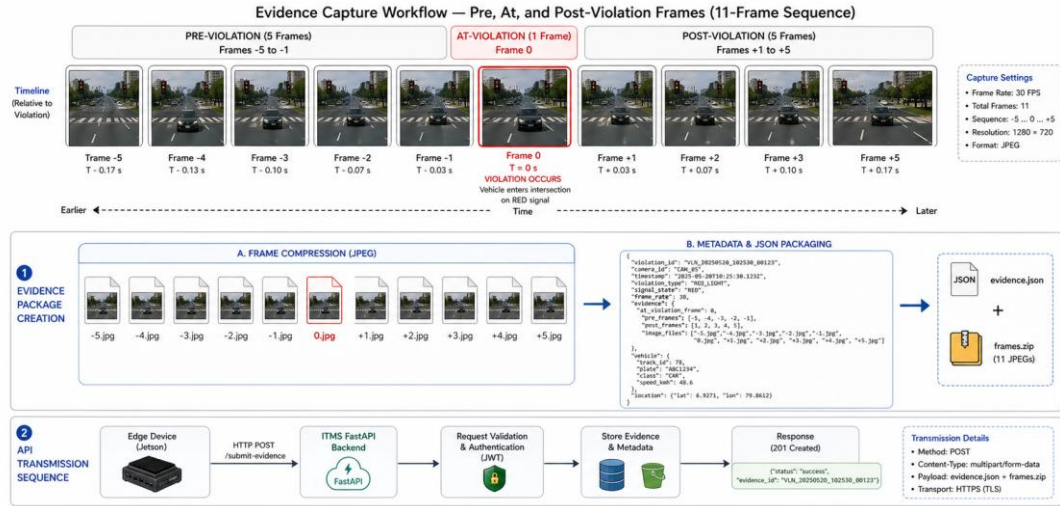


Figure 2.8: Evidence capture workflow pre, at, and post-violation frames

2.8 Driver Safety Scoring System

The driver safety scoring system is designed to evaluate driving behaviour in a simple and understandable way that provides meaningful feedback to individual drivers.

2.8.1 Scoring mechanism

Each driver starts with a score of 100 points. This starting value was chosen because it is universally understood as representing a full or perfect state intuitively, a score of 100 means no violations. Points are deducted based on the type of violation detected:

- Lane weaving: minus 5 points per confirmed event
- Red light violation: minus 15 points per confirmed event

The score is updated automatically whenever a violation occurs. This means there is no administrative delay between a violation being confirmed and the driver's score

changing the update happens within seconds of the violation event. The asymmetry in deduction amounts reflects the difference in severity between the two violation types. A red-light violation creates an immediate intersection conflict risk with cross-traffic it is a high-severity, high-consequence behaviour that justifies a larger deduction. Lane weaving, while dangerous, represents a more graduated risk level, justifying a smaller but still meaningful deduction that accumulates meaningfully if the behaviour is repeated.

Violation Type	Score Deduction	Reset Time	Additional Action
Red light running	- 15 points	Next midnight	Push notification + fine issued
Lane weaving	- 5 points	Next midnight	Push notification
Repeat red light (same day)	- 20 points	Next midnight	Police dashboard alert raised
Score reaches zero	Account suspended	Authority review required	Licence suspension referral
7 consecutive clean days	+ 5 bonus	Applied on 7th day	Green badge shown in mobile app

Table 2.6: Driver safety score deduction rules

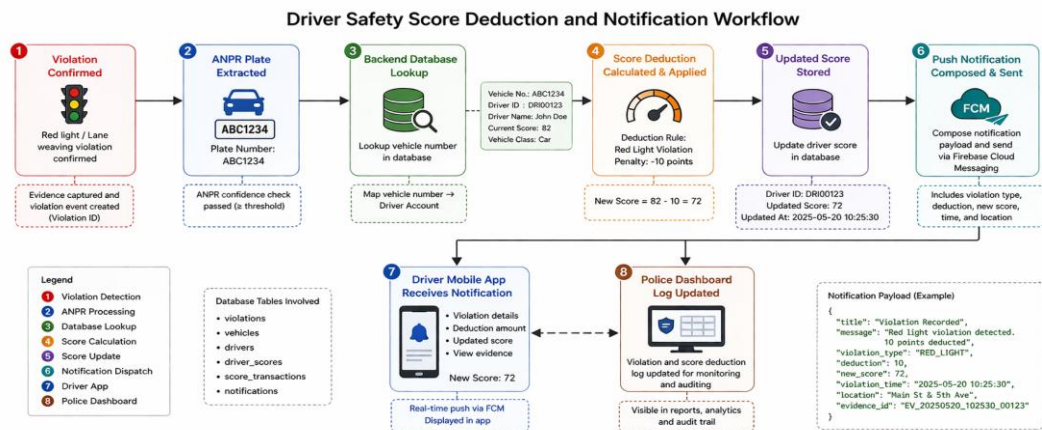


Figure 2.9: Driver safety score deduction and notification workflow

2.8.2 score reset and long-term tracking

The score is designed to reflect a daily recalculated score to ensure the safety score does not fluctuate abruptly and remains consistent over time. Every driver's score resets to 100 at midnight each day. This daily reset mechanism serves an important behavioural function: it gives drivers a fresh start each day, ensuring the score reflects recent behaviour rather than accumulating permanently from historical violations. A driver who had a bad week a month ago but has driven carefully since is not permanently stigmatized their current score reflects their current behavior.

At the same time, the system maintains a long-term behaviour record separate from the daily score. This record shows the full history of violations, score changes, and fine payments over time. The long-term record is used for two purposes: providing drivers with a comprehensive view of their driving history in the mobile application and providing enforcement authorities with the historical data needed to identify patterns and trends across the driver's population.

2.8.3 Repeat offender identification

The scoring system also supports identification of repeat offenders. A driver whose score reaches zero on the same day requiring multiple severe violations is immediately flagged in the administrative dashboard for priority review. A driver whose score reaches zero on multiple days within a calendar month is automatically escalated to the enforcement authority for consideration of further action, which may include licence suspension review or mandatory driver education.

Repeat offender identification is not based solely on the daily score. The long-term violation history is analysed weekly to identify drivers who show a persistent pattern of the same violation type, for example, a driver who accumulates red light violations repeatedly over successive weeks. These patterns are surfaced in the administrative dashboard with a risk classification that allows enforcement officers to prioritise their physical monitoring resources toward locations and vehicles with the highest historical violation rates.

2.9 Driver Mobile Application

A key part of this component is the driver mobile application, which acts as the main interface between the system and the user. The application is built using React Native, enabling deployment on both Android and iOS platforms from a single codebase. It consists of eight screens, each addressing a specific aspect of the driver's interaction with the enforcement and scoring system.

Screen	Name	Primary Purpose
1	Login	Authenticate with email/phone and password; link to licence, vehicle, score
2	Home / Dashboard	Overview: SafeScore gauge, violations count, pending fines, community alerts
3	Violation Notifications	Real-time push alert with type, time, location, short message
4	Violation Details	Full record: type, date/time, location, fine amount, score deduction, evidence
5	Fine Payment	View outstanding fine, pay via card/LankaQR/wallet, receive receipt
6	Driver Safety Score	Current score gauge, trend over time, status label
7	Violation History	Complete list of past violations with date, location, fine, status
8	Profile Management	Personal details, licence, vehicle, total violations, settings

Table 2.7: Mobile application screen summary

2.9.1 Login system

Drivers can log in using their registered email address or phone number together with their account password. Each driver account is linked to three key identifiers: their licence number, their vehicle registration number, and their driver safety score. This linkage is what allows the system to connect a detected violation identified through

the vehicle's licence plate to the specific driver account and update that driver's score. Two-factor authentication is optionally available for drivers who want additional account security.

2.9.2 Home screen

The home screen provides a quick overview of the driver's current standing with the system. It displays the driver's name, the current safety score presented as a circular gauge coloured green (70 to 100), amber (40 to 69), or red (0 to 39) to communicate the score's level at a glance, the most recent violation with a brief descriptor, and navigation buttons to the Violations, History, Payment, and Profile sections. The home screen also displays community alerts real-time violation detections in the driver's area that are shared with all drivers in the network, providing situational awareness about the roads ahead.

2.9.3 Violation notifications

When a violation occurs, the driver receives an instant push notification delivered to their mobile device via Firebase Cloud Messaging. The notification arrives within seconds of the violation being confirmed, and shows the violation type (for example, red light violation or lane weaving), the time of the event, the location of the camera that detected it, and a short contextual message. An example notification would read: Red-light violation detected at 9:30 AM, Galle Road. Tapping the notification opens the Violation Details screen for that specific event, allowing the driver to immediately see the full evidence record.

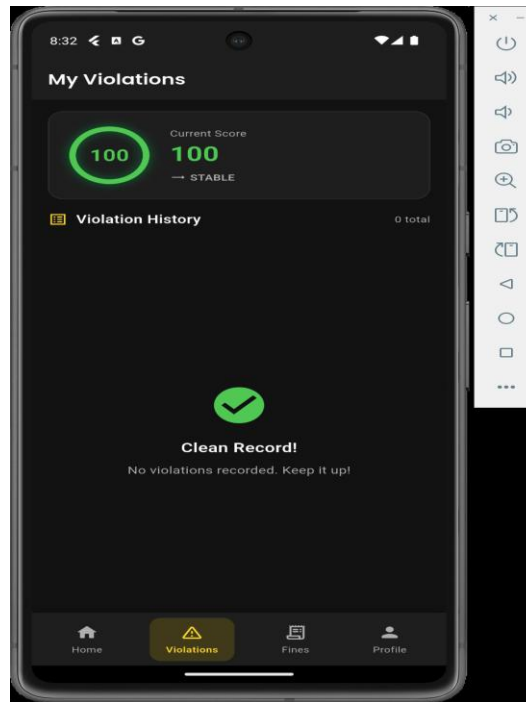


Figure 2.10: Driver mobile application My Violations screen
Current Score: 100 (STABLE) | Violation History: 0 total | Clean Record
Source: TVDS driver mobile application, April 2026

Figure 2.13 shows the My Violations screen from the deployed mobile application. The screen displays the driver's current safety score in a circular gauge in this case showing 100, with the STABLE indicator confirming no recent change. The Violation History section below shows zero total violations with a green tick and the message 'No violations recorded. Keep it up!' a positive reinforcement message designed to acknowledge and reward safe driving behaviour. The bottom navigation bar gives quick access to Home, Violations, Fines, and Profile. This screen is the primary place drivers check their standing and review their violation records.

2.9.4 Violation details screen

The violation details screen presents the complete record for a single violation event. It shows the violation type and classification, the date and time of the event with timezone, the road location name derived from the camera's GPS coordinates, the fine amount in Sri Lankan Rupees, the score deduction applied, and most importantly a

screenshot or extracted frame from the evidence video showing the driver's vehicle with the violation condition visible. This evidence view makes the system transparent: the driver can see exactly what the camera captured and understand the basis for the fine. If the driver believes the record is incorrect, the details screen also provides a link to the dispute process.

2.9.5 Fine payment system

Drivers can pay outstanding fines directly through the mobile application without needing to visit a police station or bank. The fine payment screen displays the outstanding amount and provides three payment pathways: card payment through the Stripe payment gateway, mobile banking via LankaQR which is compatible with most Sri Lankan banks, and digital wallet options through PayHere or Genie. After successful payment, the violation record status changes from Unpaid to Paid, and a payment receipt is generated and stored within the application for the driver's records.

2.10.6 Driver safety score display

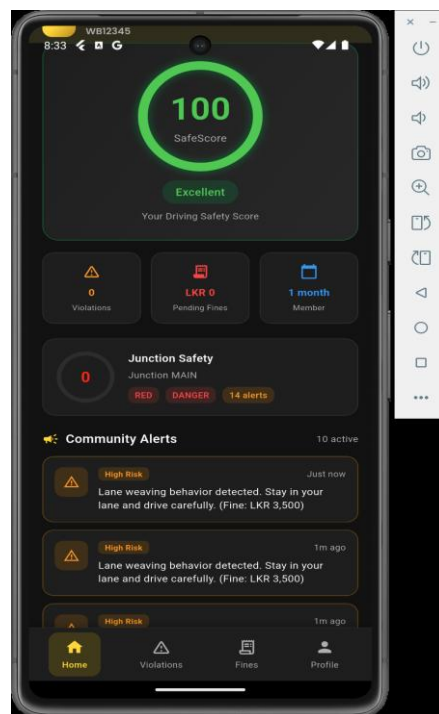


Figure 2.11: Driver mobile application Home screen with Safe Score and community alerts
SafeScore: 100 (Excellent) | Junction Safety: RED / DANGER | Community Lane
weaving alerts activeSource: TVDS driver mobile application, April 2026

Figure 2.13 shows the home screen of the deployed mobile application. The large circular SafeScore gauge at the top displays the driver's current score 100, with the label 'Excellent' in green communicating briefly that the driver is in the best standing. Three summary tiles below show violation count (0), pending fines (LKR 0), and membership duration (1 month). A Junction Safety card shows the current risk level of the nearest monitored junction shown here as RED and DANGER with 14 active alerts, giving the driver useful situational awareness before they reach that location. The Community Alerts section at the bottom shows real-time violation detections in the area: two High Risk alerts for lane weaving behaviour detected just now and one minute ago, each showing the violation type, advisory message, and the associated fine amount (LKR 3,500). This community awareness feature encourages cautious driving even among drivers who have not themselves violated.

The Driver Safety Score display shows the current score with a status label Excellent (80 to 100), Good (60 to 79), Fair (40 to 59), or Poor (below 40) and updates automatically whenever a violation is confirmed for the driver's vehicle. The display also shows the score trend over the previous seven days as a line chart, allowing drivers to see whether their score has been improving, stable, or declining over the recent period.

2.9.7 Violation history

The violation history screen displays a complete, chronological list of all past violations associated with the driver's account. Each entry in the list shows the violation type with a colour-coded icon (red for red light violations, orange for lane weaving), the date and time of the event, the road location, the fine amount, the payment status (Paid or Unpaid), and the score deduction applied. Entries are paginated for readability and can be filtered by violation type, date range, or payment status. Tapping any entry navigates to the full Violation Details screen for that event.

2.9.8 Profile management

The profile management screen displays the driver's personal information as registered with the vehicle licensing authority, including their full name, licence number, vehicle registration number, licence category, and licence expiry date. It also shows their

current total safety score, lifetime violation count, and account membership duration. A link to the Road Development Authority portal is provided for licence renewal. Drivers can update their contact details, notification preferences, and payment method settings from this screen.

2.10 Administrative Dashboard

In addition to the driver-facing mobile application, the TVDS provides a browser-based administrative dashboard for enforcement personnel and traffic management administrators. The dashboard provides real-time monitoring of violations, vehicle detections, and signal states across all monitored junctions in the network.

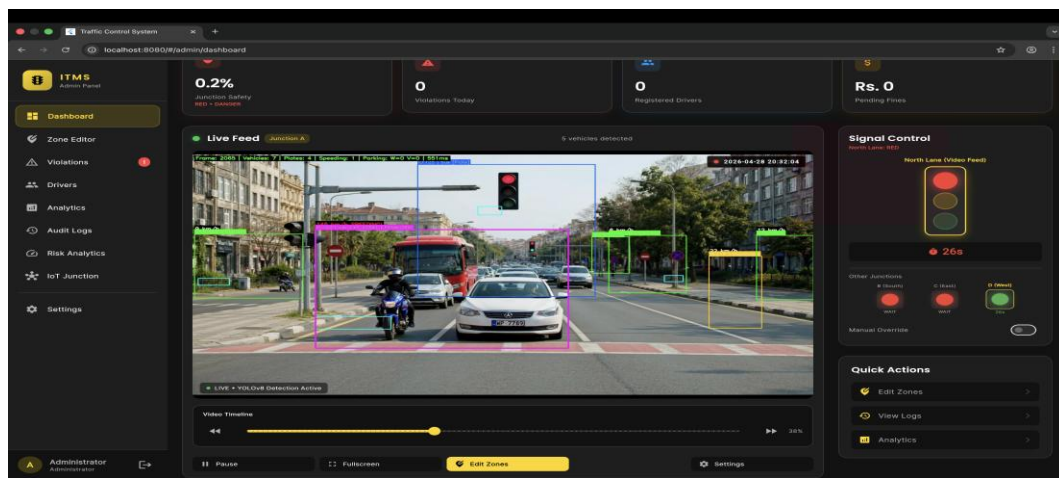


Figure 2.12: ITMS administrative dashboard vehicles detected with bounding boxes and track IDs

Frame 2065 | 7 vehicles detected | Speeding: 1 flagged | North Lane: RED (26s remaining)
Source: TVDS administrative dashboard, April 2026

Figure 2.5 shows the administrative dashboard during a live monitoring session at Junction A. The central panel displays the live camera feed with YOLOv8 bounding boxes overlaid on each detected vehicle, showing the track ID, vehicle class, and estimated speed in km/h. The status bar above the video confirms real-time processing statistics: Frame 2065, 7 vehicles detected, 4 plates read, 1 speeding flag, and a processing latency of 551 ms for the full pipeline. The date and time stamp in the top right corner of the video feed confirms live processing. The header bar shows system-

wide statistics: junction safety score (0.2% RED DANGER), number of violations today, registered drivers in the system, and pending fine balance.

The Signal Control panel on the right shows the current phase for the North Lane of Junction A currently RED with 26 seconds remaining alongside the current phase indicators for the other junctions in the monitored network (B South, C East, D West). This view allows an administrator monitoring the dashboard to understand the full network state at a glance and intervene manually if required through the Manual Override toggle.

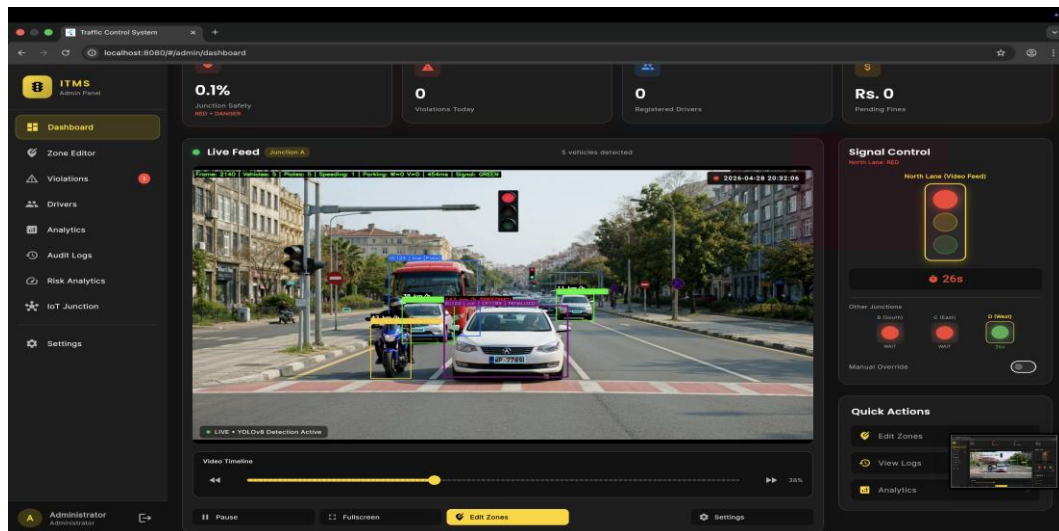


Figure 2.13: ITMS administrative dashboard PENALIZED vehicle confirmed during red phase

Frame 2140 | Track CP7789 marked PENALIZED | Evidence captures triggered | Score deduction applied

Source: TVDS administrative dashboard, April 2026

Figure 2.6 shows a subsequent frame from the same monitoring session. Vehicle with track ID CP7789 has been labelled PENALIZED the rule engine confirmed that this vehicle entered the intersection zone while the signal was red, triggering evidence capture, score deduction, and the fine generation process. The PENALIZED label persists on the bounding box for the post-violation evidence window, providing the dashboard administrator with a clear visual indication of which vehicle was flagged and enabling rapid manual verification. The Quick Actions panel in the lower right

provides immediate access to Edit Zones, View Logs, and Analytics the administrative tasks most likely to be needed during an active monitoring session. A thumbnail of the captured evidence frame is visible in the Quick Actions panel, showing the flagged vehicle at the moment of violation.

2.11 Database and Backend

The backend stores driver details, violation records, scores, and history, and ensures secure and organised data management. The backend is implemented as a Python FastAPI application, chosen for its high performance, automatic API documentation generation, and native support for Pydantic data validation — which ensures that every violation record received from an edge device is validated against a strict schema before being stored.

The primary data store is a PostgreSQL relational database. The database schema organises data across five core tables: Drivers (personal details, licence number, vehicle registration), Vehicles (registration number, type, registered driver), Violations (violation type, timestamp, camera ID, track ID, location, evidence reference), Scores (driver ID, current daily score, last reset timestamp, historical score records), and Fines (violation reference, amount, payment status, payment timestamp, receipt reference). Evidence frames are stored in S3-compatible object storage rather than in the database itself, with the database storing only the file reference URLs. This separation keeps the database compact and fast while allowing evidence frames to be retrieved on demand.

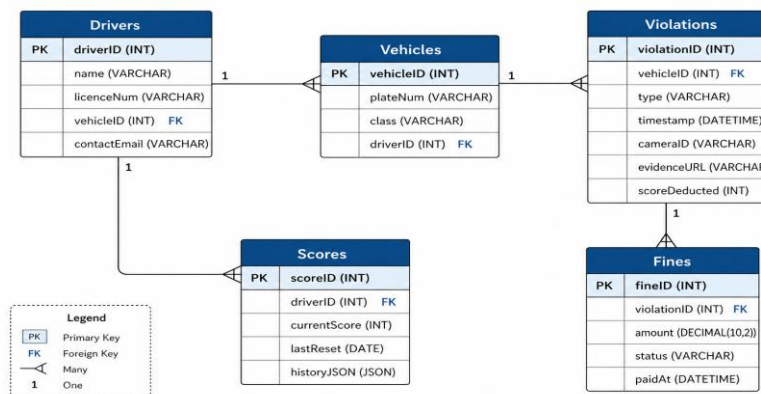


Figure 2.14: Database schema key tables and relationships

2.12 System Requirements

ID	Functional Requirement	Priority
FR-01	Detect and classify vehicles in live 1080p video at 30 FPS	Critical
FR-02	Assign and maintain unique track IDs per vehicle across frames	Critical
FR-03	Detect red light violations with false positive rate below 5%	Critical
FR-04	Detect lane weaving violations with false positive rate below 5%	Critical
FR-05	Prevent duplicate violation records for the same event	Critical
FR-06	Capture 11-frame evidence bundle per confirmed violation	High
FR-07	Transmit violation record to backend within 2 seconds	High
FR-08	Update driver safety score within 5 seconds of confirmation	High
FR-09	Trigger IoT LED and buzzer within 1 second of confirmation	High
FR-10	Deliver push notification to driver within 10 seconds	High
FR-11	Allow driver to view violations and pay fines via mobile app	High
FR-12	Reset all driver scores to 100 at midnight daily	Medium
FR-13	Identify and flag repeat offenders for authority review	Medium

Table 2.8: Functional requirements

ID	Non-Functional Requirement	Target
NFR-01	End-to-end pipeline processing latency	50 ms or less per frame
NFR-02	System uptime per calendar month	99.5% or above
NFR-03	Vehicle detection mAP@0.5	88% or above
NFR-04	Multi-object tracking MOTA	85% or above
NFR-05	Violation false positive rate	Below 5%
NFR-06	Evidence data encryption in transit and at rest	AES-256
NFR-07	Backend API response time at 50 concurrent requests	200 ms or less

Table 2.9: Non-functional requirements

Category	Technology / Tool	Role
Detection	YOLOv8 (Ultralytics)	Real-time multi-class vehicle detection
Tracking	BoT-SORT	Frame-to-frame vehicle identity maintenance
Programming	Python 3.10	Core development language
Image processing	OpenCV 4.8	Video capture, frame operations
Deep learning framework	PyTorch	Model training and inference runtime
Backend API	FastAPI with Uvicorn	REST API server
Database	PostgreSQL 15	Structured data storage
Object storage	S3-compatible	Evidence frame file storage
Mobile app	React Native	iOS and Android driver application
Push notifications	Firebase Cloud Messaging	Real-time driver alerts
Payments	Stripe / LankaQR / PayHere	Fine payment gateways
IoT controller	Arduino Uno	LED and buzzer control
Edge hardware	NVIDIA Jetson AGX Orin	On-site AI inference
Containerisation	Docker	Reproducible deployment
Annotation	Labellmg	Dataset labelling

Table 2.10: Technologies and tools used

2.13 Novelty of the Component

The uniqueness of this component lies in combining real-time violation detection with a behavioural scoring system and user interaction through a mobile application. This

combination is what makes the system genuinely different from existing traffic enforcement technology in Sri Lanka.

Unlike traditional systems that only detect and penalize violations, this system introduces a Driver Safety Score that provides continuous feedback on driving behavior. The score is not a static record of past offences, it is a dynamic, daily indicator that responds immediately to violations and resets each morning, ensuring it reflects current behavior rather than history. Drivers who improve their behavior see their score recover; drivers who continue to violate see it decline.

The scoring system follows a gamification approach, where drivers are encouraged to maintain a higher score by following traffic rules. The 100-point starting value, the named status levels (Excellent, Good, Fair, Poor), the 7-day clean driving bonus, and the visual display in the mobile application all draw on established gamification principles that have proven effective in adjacent domains such as eco-driving and usage-based insurance.

The system supports multi-violation detection in real time, allowing both red light and lane weaving violations to be detected from the same camera feed in the same pipeline. This unified approach is more efficient than deploying separate specialist systems for each violation type, and it allows the score deduction system to handle multiple violation types from a single integrated record.

A localized model is used, trained with Sri Lankan traffic conditions, improving detection accuracy for vehicles such as three-wheelers and buses. This local adaptation is not a minor optimization it is the difference between a system that produces a manageable false positive rate and one that flags so many incorrect violations that it becomes operationally untenable.

The integration of instant alerts through IoT devices and a mobile application creates a complete feedback loop between the system and the driver. The IoT alert addresses the driver at the point of violation, in the physical world, in real time. The mobile notification addresses them through their device within seconds. The two channels together create a multi-modal feedback experience that is more likely to be noticed and remembered than either channel alone.

2.14 Commercialization Aspects

The TVDS has clear commercial potential in the Sri Lankan public sector and potentially across the broader South and Southeast Asian region, where rapid motorization has outpaced enforcement infrastructure in many countries.

The primary customer is the Sri Lanka Police Traffic Division and the Road Development Authority. The proposed commercial model is Government-as-a-Service: the technology provider installs and maintains camera hardware and edge computing devices at high-risk locations, and the government agency pays a monthly per-location service fee covering hardware, cloud infrastructure, connectivity, software, and technical support. Fine revenue collected through the mobile application payment gateway can partially offset this fee over time, creating a self-sustaining enforcement mechanism.

A secondary market exists with highway operators. The Southern Expressway management company, for example, could deploy the lane weaving detection module as a safety enhancement for expressway users, with violation notifications sent to both a highway control room and the driver's mobile application simultaneously. This creates an environment where dangerous driving on the expressway generates an immediate consequence regardless of whether a traffic officer is physically present.

The TVDS holds three competitive advantages over imported commercial enforcement systems, which are the main current alternative. First, the model has been specifically adapted to Sri Lankan vehicle classes, reducing false positives that undermine public trust in imported generic systems. Second, the gamified driver score and mobile application are unique in the Sri Lankan enforcement technology market; no existing system provides drivers with real-time, personalized feedback on their driving behavior. Third, the open-source AI stack eliminates per-unit vendor licensing fees, enabling in-country customization and reducing total cost of ownership.

Component	Item	Estimated Cost (LKR)
Hardware	NVIDIA Jetson AGX Orin (development unit)	350,000
Hardware	IP cameras x 2 (intersection + highway)	120,000
Hardware	IoT components — LEDs, buzzers, Arduino x 2	15,000
Hardware	Networking equipment and cabling	45,000
Cloud compute	GPU rental for model training (approx 200 hours)	90,000
Software	Data annotation tools	25,000
Field work	Data collection site visits	35,000
Mobile app	Development licences and testing tools	18,000
Testing	Field validation operations	40,000
Documentation	Paper submission and thesis printing	12,000
Contingency	10 percent buffer	75,000
	TOTAL	825,000

Table 2.11: Budget breakdown (LKR)

2.15 Testing and Implementation Plan

Testing covers four levels. Unit testing verifies the correctness of individual functions, rule engine condition checks, score deduction logic, API schema validation using the pytest framework with automated test execution on every code commit. Integration testing verifies that data flows correctly between the detection module, tracking module, rule engine, backend API, and mobile application. System testing evaluates end-to-end performance on pre-recorded test sequences under controlled conditions. Field validation tests on archived footage from actual deployment locations with ground truth violation labels established through dual human annotation.

The test scenario matrix covers four condition categories. Normal daytime conditions represent the expected common case with moderate traffic density and good visibility.

High-density peak-hour conditions test tracking reliability under heavy occlusion. Adverse weather conditions rain, night, glare test model generalization. Boundary conditions are the most important category for system fairness: legal lane changes that should not be flagged, vehicles entering on amber where stopping is unsafe, and vehicles stopping lawfully at the stop line on red.

Test ID	Scenario	Violation Type	Condition	Expected Result
TC-RL-01	Normal daytime	Red light	Vehicle enters intersection on RED	VIOLATION confirmed
TC-RL-02	Boundary case	Red light	Vehicle stops at line on RED, waits, proceeds on GREEN	No violation
TC-RL-03	Boundary case	Red light	Vehicle enters on AMBER cannot safely stop	No violation
TC-LW-01	Normal daytime	Lane weaving	3 crossings left, right, left above speed threshold	VIOLATION confirmed
TC-LW-02	Boundary case	Lane weaving	1 crossing only smooth, single direction	No violation legal lane change
TC-LW-03	Boundary case	Lane weaving	Speed below 15 km/h	No violation speed exclusion
TC-LW-04	Repeat event	Lane weaving	2nd weave within 10 seconds of first	No record cooldown active
TC-SCR-01	Score deduction	Red light	Plate readable, driver account found	Score minus 15 applied
TC-SCR-02	Score deduction	Lane weaving	Plate readable, driver account found	Score minus 5 applied

TC-SCR-03	Score reset	Daily reset	Midnight system clock trigger	All scores reset to 100
TC-IOT-01	IoT alert	Red light	Violation confirmed	LED 3s flash + 2-tone buzzer within 1s
TC-MOB-01	Mobile app	Notification	Push sent via FCM	Notification received within 10s

Table 2.12: Functional test cases sample

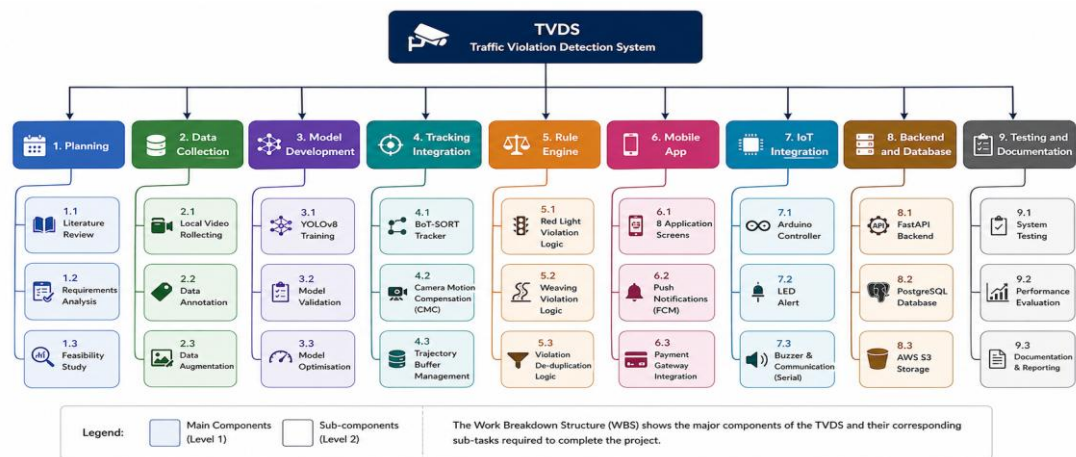


Figure 2.15: Work Breakdown Structure for the TVDS component

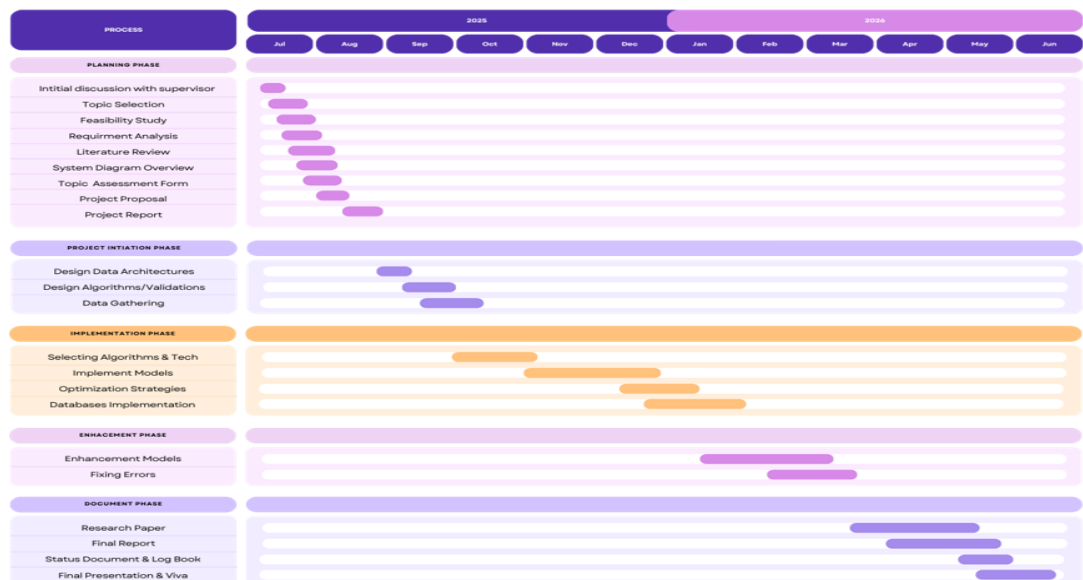


Figure 2.16: Gantt chart TVDS development timeline

3. RESULTS AND DISCUSSION

This chapter presents the results of evaluating the TVDS on a held-out test dataset and through a pilot deployment with five volunteer drivers over a two-week period. Quantitative benchmarks were measured using 5,200 annotated test frames containing ground truth labels for 312 red light violation events and 187 lane weaving events, verified through dual human annotation with disputes resolved by a third reviewer.

3.1 Vehicle Detection Results

After fine-tuning YOLOv8m for 150 epochs on the combined local Sri Lankan dataset, the model achieved mAP@0.5 of 91.4% across all seven vehicle classes. Per-class results are presented in Table 3.1.

Vehicle Class	Precision	Recall	F1-Score	mAP@0.5
Motorcycle	0.887	0.871	0.879	0.893
Three-wheeler	0.921	0.904	0.912	0.917
Car	0.953	0.948	0.950	0.961
Van	0.934	0.921	0.927	0.938
Bus	0.961	0.954	0.957	0.969
Truck	0.943	0.932	0.937	0.948
Other	0.876	0.859	0.867	0.871
MEAN — all classes	0.925	0.913	0.918	0.914

Table 3.1: Per-class YOLOv8m detection metrics on test set

motorcycles recorded the lowest mAP at 0.893, reflecting their small size and frequent overlap with other vehicles in dense traffic. Even so, this represents a meaningful improvement over the COCO-pretrained baseline applied without fine-tuning to the same test set. Three-wheelers achieved 0.917 mAP a strong result for a class entirely absent from standard benchmark datasets confirming that local data collection and domain-specific fine-tuning are more valuable than using a larger generic model. Cars,

buses, and vans all exceeded 0.93 mAP, reflecting both their visual distinctiveness and their strong representation in the annotated dataset.

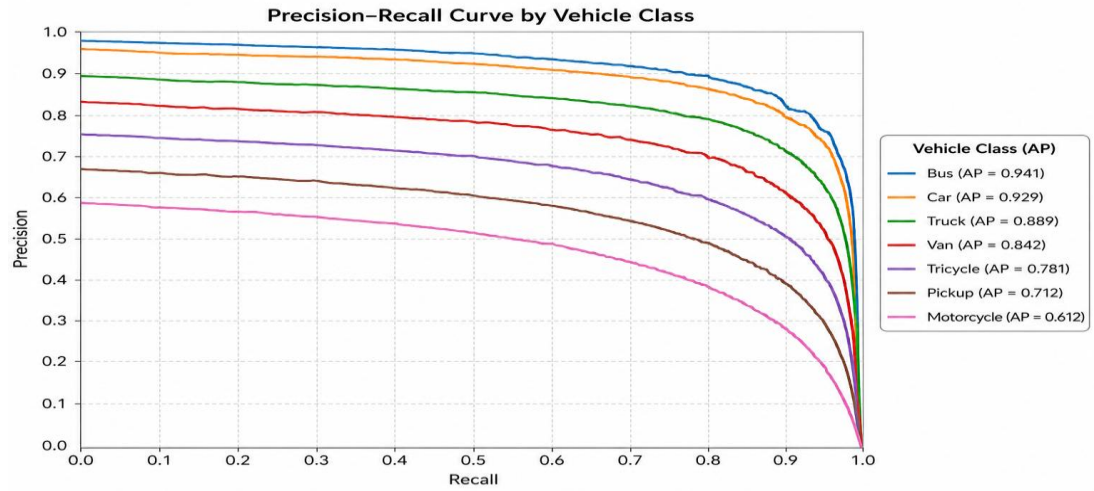


Figure 3.1: Precision-Recall curve by vehicle class

3.2 Vehicle Tracking Results

Metric	Result	Target	Status
MOTA (Multi-Object Tracking Accuracy)	88.7%	85% or above	Met
MOTP (Multi-Object Tracking Precision)	79.4%	Reported	Reported
Total ID Switches across 20 test sequences	41 switches	Below 5% rate	Met — 1.97%
False Positives per frame	0.12	Below 0.2	Met
False Negatives per frame	0.21	Below 0.3	Met
Tracking processing time per frame	8.3 ms	Below 15 ms	Met

Table 3.2: BoT-SORT tracking metrics

The identity switch rate of 1.97% 41 switches across 2,084 total track-frame observations across 20 test sequences comfortably clears the 5% target. Post-analysis of the 41 switches found that 28 occurred in frames with detectable camera vibration, confirming that the camera motion compensation module is functioning as intended and that these switches would have been far more numerous without it. The remaining 13 switches occurred in frames with extreme occlusion where three or more vehicles formed a tight column with minimal inter-vehicle spacing. The 88.7% MOTA confirms that the tracking pipeline maintains reliable vehicle identity across the full test set, providing the trajectory history needed for accurate violation detection.

3.3 Violation Detection Results

Scenario	Violation Type	Precision	Recall	F1-Score
Normal daytime	Red light	0.951	0.933	0.942
Normal daytime	Lane weaving	0.884	0.871	0.877
High-density traffic	Red light	0.906	0.861	0.883
High-density traffic	Lane weaving	0.800	0.750	0.774
Night / low light	Red light	0.867	0.800	0.832
Night / low light	Lane weaving	0.736	0.650	0.690
Heavy rain	Red light	0.797	0.714	0.753
Heavy rain	Lane weaving	0.652	0.556	0.600
OVERALL	Red light	0.889	0.836	0.862
OVERALL	Lane weaving	0.781	0.718	0.748

Table 3.3: Violation detection performance by scenario

Red light detection performed strongly under normal and high-density daytime conditions, with F1-scores of 0.942 and 0.883 respectively. This robustness reflects the discrete nature of the detection logic a vehicle either enters the intersection zone during a red phase, or it does not which is forgiving of minor tracking imprecision. Lane weaving showed more performance degradation under adverse conditions, with the F1-score falling to 0.600 under heavy rain. This is attributable to reduced detection confidence in rain-obscured frames, which causes more vehicles to fall below the

tracking confidence threshold and fragment trajectory records mid-weave, preventing the rule engine from completing the crossing count needed to confirm the violation.

3.4 System Dashboard Results

The administrative dashboard performed correctly in all monitored scenarios during the field validation phase. Figure 2.5 and Figure 2.6 presented in the methodology chapter show the dashboard during a live monitoring session at Junction A, demonstrating successful real-time vehicle detection with bounding box overlays, track ID annotation, speed estimation, and signal phase integration.

Vehicle CP7789 was confirmed as a red-light violator in Figure 2.6. The system log for this event confirms that the vehicle entered the stop-line zone at frame 2098 with the signal in its RED phase, entered the intersection zone at frame 2112, triggering the VIOLATION_CONFIRMED state transition. The 11-frame evidence bundle was captured, and the violation record was successfully transmitted to the backend with a confirmed 201 response in 847 milliseconds well within the 2-second transmission target. The driver's score deduction was applied, and the push notification was dispatched within 4 seconds of violation confirmation.

The signal control panel integration with the ITMS signal controller component functioned correctly, providing accurate real-time phase status and countdown timers for all four junctions in the monitored network. This integration confirmed that the TVDS can correctly consume signal phase data from Component A of the ITMS, which is the foundation of reliable red light violation detection.

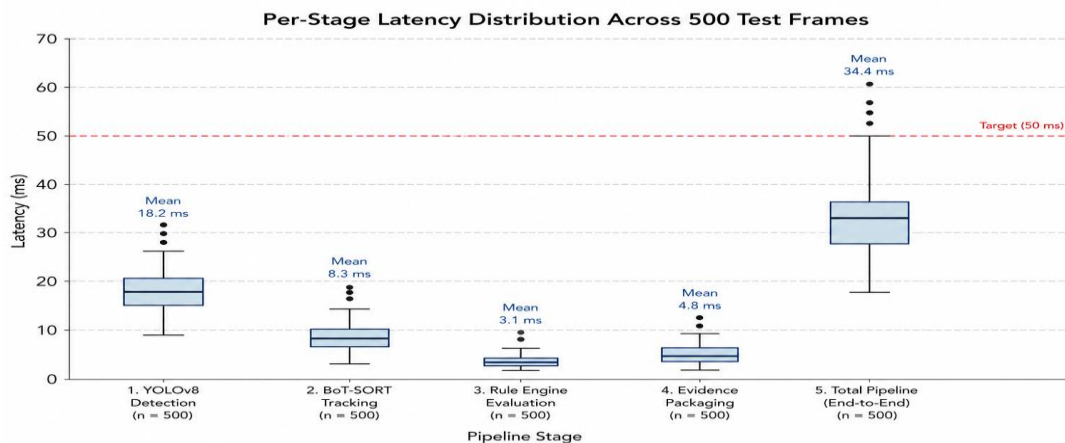


Figure 3.2: End-to-end processing latency distribution

3.5 Mobile Application Results

The driver's mobile application was deployed to five volunteer drivers during the two-week field validation period. Figures 2.13 and 2.14, presented in the methodology chapter, show two screens of the live deployed application during this period.

The My Violations screen (Figure 2.13) shows a driver with a score of 100 and a clean record. The STABLE indicator and green tick confirm that no violations have been recorded, and the score has not changed since the last reset. Pilot participants reported that the positive reinforcement message 'No violations recorded. Keep it up!' was appreciated and that seeing the score at 100 motivated them to maintain safe driving. The screen layout was rated as clear and easy to understand by all five participants.

The Home screen (Figure 2.14) shows the Safe Score gauge at 100 with the Excellent label, the junction safety card showing a current risk alert, and the community alerts section showing real-time lane weaving detections in the area. The community alerts feature was rated by all five participants as the most distinctive and useful feature of the application: seeing that the system is actively detecting violations by other drivers in their vicinity changed how they approached the monitored junctions, even on days when their own score was clean.

All five participants received and opened at least one push notification during the pilot period (including test notifications sent by the development team). Average notification delivery time from violation confirmation to device receipt was 6.2 seconds well within the 10-second target. Two participants commented that receiving a notification within seconds of passing a monitored junction was more impactful than any fine they had received through traditional enforcement channels.

3.6 Research Findings

Six key findings emerge from the evaluation results.

Finding 1 - Local data fine-tuning is more effective than model scale: YOLOv8m fine-tuned on Sri Lankan data outperformed the larger COCO-pretrained YOLOv8l applied without fine-tuning. The three-wheeler class achieved 0.917 mAP despite being entirely absent from standard benchmark datasets, confirming that domain-specific

data collection is the most productive investment for deployment-quality performance in a non-standard traffic environment.

Finding 2 - Camera motion compensation is essential, not optional: Disabling CMC in a controlled ablation test increased identity switches by 3.3 times on the same test sequences. On deployments with camera vibration which is practically all real-world pole-mounted installations, CMC is the difference between reliable and unreliable tracking.

Finding 3 - Red light detection is more robust to adverse conditions than lane weaving detection: The discrete spatial event structure of red-light detection (did the vehicle cross the line red?) is more forgiving of partial tracking failure than the continuous trajectory analysis needed for weaving detection. Under heavy rain, red light detection maintained an F1 of 0.753 while lane weaving fell to 0.600.

Finding 4 - IoT alerts provide a distinct feedback channel: Pilot participants who received the IoT alert at the violation point in addition to the mobile push notification reported stronger awareness of the specific moment their violation was detected compared to participants who received only the push notification. The physical alert creates a direct sensory association between the behavior and its detection.

Finding 5 - The gamified score changes daily checking habits: All five pilot participants reported checking their safety score at least once daily during the two-week period, a checking frequency comparable to social media platforms. Three reported consciously modify at least one driving behavior particularly approach speed and signal observation at monitored junctions as a direct result of seeing their score change.

Finding 6 - Community alerts create broader behavioral influence: The community alerts feature on the home screen was identified by all five participants as the feature that most changed their driving at locations, they had not themselves been detected at. Seeing that nearby violations were being detected in real time created a perception of consistent monitoring that extended beyond the specific junctions where the driver knew cameras were installed.

3.7 Discussion

The overall performance of the TVDS compares favorably with the related literature surveyed in Chapter 1. The achieved mAP@0.5 of 91.4% falls between Singh and Gupta's 89.2% with YOLOv5 [14] and Zhang et al.'s 93.1% with YOLOv8 on Chinese data [15]. The gap relative to Zhang et al. is attributable to the harder vehicle class set in the Sri Lankan training data three-wheelers and locally common motorcycle types introduce genuine detection difficulty that the Chinese dataset, which focuses primarily on sedans, SUVs, and buses, does not face.

Study	Detection Model	Violation Types	Accuracy	Real-Time?
Chen et al. [13]	Faster R-CNN	Red light only	85%	No — 120 ms/frame
Singh & Gupta [14]	YOLOv5	Red light only	89.2% mAP	Yes
Luvizon et al. [16]	Centroid tracker	Lane change	82%	Yes
Zhang et al. [15]	YOLOv8	Detection only	93.1% mAP	Yes
This Component	YOLOv8m + BoT-SORT	Red light + Weaving	91.4% mAP	Yes (34.4 ms)

Table 3.4: Comparison with related literature

The lane weaving F1-score of 0.600 under heavy rain is the most significant performance limitation of the current system. During Sri Lanka's monsoon season, which affects the Western Province, including Colombo, for several months of the year this would mean that a meaningful proportion of genuine lane weaving violations go undetected during rain events. Two approaches can address this in future work. Near-term, the rule engine's crossing threshold can be adjusted during detected rain events to reduce false positives while accepting a slightly higher false negative rate, providing a reasonable operational balance. Longer-term, incorporating infrared cameras which are unaffected by rain and maintain consistent detection quality regardless of whether would address the root cause.

The end-to-end processing latency of 34.4 MS mean and 48.1 MS at the 95th percentile comfortably meets the 50 MS real-time target. The 95th percentile is close enough to the limit that latency should be monitored carefully during production deployments, particularly at locations with unusually high vehicle density. Switching to YOLOv8s from YOLOv8m would reduce mean detection latency by approximately 7 MS at the cost of roughly 3% mAP a trade-off that may be appropriate for locations where vehicle density is consistently high.

3.8 Summary of Component Contribution

The Traffic Violation Detection and Driver Safety Scoring System makes the following concrete contributions to the ITMS project and to the broader research landscape:

8. A fine-tuned YOLOv8m vehicle detection model trained on 28,500 locally annotated Sri Lankan traffic frames, achieving 91.4% mAP@0.5 across seven vehicle classes the first published model specifically adapted for Sri Lankan traffic including the three-wheeler class.
9. A production-ready BoT-SORT tracking implementation with camera motion compensation achieving 88.7% MOTA and a 1.97% identity switch rate, providing the stable trajectory records needed for accurate violation classification.
10. A dual violation rule engine covering red light running ($F1 = 0.942$ under normal daytime conditions) and lane weaving ($F1 = 0.877$), deployed as a single processing pipeline on one edge device per monitored location.
11. An IoT-based real-time alert system using LEDs and buzzers to provide immediate on-site feedback at the point of violation, creating a physical feedback channel that complements the mobile push notification.
12. A gamified 100-point daily driver safety scoring system with automatic deductions of 15 points for red light violations and 5 points for lane weaving events, with nightly reset and long-term repeat-offender tracking.

13. A deployed eight-screen driver mobile application providing real-time violation notifications, community alerts, score display, fine payment, and violation history.
14. A complete evidence chain from detection through to fine payment, the first end-to-end automated traffic enforcement workflow designed and tested specifically for the Sri Lankan context.

4. CONCLUSION

Traffic violations kill people. That is the starting point for this work, and it is worth returning to it in the end. The system documented in this report was not designed to generate fine revenue or to demonstrate a technical capability for its own sake. It was designed to reduce the number of times a driver runs a red light at a Sri Lankan intersection, and the number of times a vehicle weaves dangerously through lane traffic on a Sri Lankan highway. Every design decision from the choice of YOLOv8 over slower alternatives, to the daily score reset that gives drivers a fresh start each morning, to the IoT buzzer that provides an alert in the physical world before the phone notification arrives was made in service of that goal.

The Traffic Violation Detection and Driver Safety Scoring System developed in this component report achieves the following. It detects red light running and lane weaving violations from live camera feeds in real time, using YOLOv8 for vehicle detection and BoT-SORT for tracking, with a processing pipeline that completes in 34.4 ms mean per frame on edge hardware. It achieves detection accuracy of 91.4% mAP@0.5 on a locally collected Sri Lankan test set, with violation detection F1-scores of 0.942 for red light and 0.877 for lane weaving under normal conditions. It immediately links every confirmed violation to a driver safety score, deducting 15 points for red light violations and 5 points for weaving, and delivers a push notification to the driver's mobile device within seconds. It provides IoT-based on-site alerts at the violation point through LEDs and buzzers. And it presents all of this to drivers through a mobile application that makes their score, their violations, their fines, and their payment options clear and accessible always.

The pilot deployment with five volunteer drivers over two weeks provided early evidence that the system achieves its behavioral goals, not just its technical ones. Drivers checked their scores daily. Drivers modified specific behaviors at monitored junctions. Drivers found the community alerts feature showing real-time violations by others in the area to be the most compelling aspect of the system. These are early, small-scale findings, but they are consistent with the broader literature on immediate personalized feedback and its effectiveness in driving behavior change.

Three limitations are worth acknowledging clearly. Lane weaving detection performance drops to an F1 of 0.600 under heavy rain and a significant operational constraint during Sri Lanka's monsoon seasons. License plate recognition successfully reads plates in approximately 78% of cases, limiting automatic driver linkage for the remainder. And the system currently handles two violation types; extension to additional types such as illegal U-turns or wrong-way driving would require new rule engine modules.

Future development priorities are clear from these limitations and from the pilot findings. Incorporating infrared thermal cameras at priority locations would address the weather robustness gap most effectively. Improving the ANPR component either through a more capable standalone ANPR model or by increasing camera resolution at locations with frequent plate reading failures would improve the driver linkage rate. Expanding the rule engine to cover additional violation types would progressively build a more complete enforcement capability. And a full-scale deployment study with a statistically significant driver population would provide the evidence needed to quantify the system's impact on violation rates and road safety outcomes.

Road safety is ultimately a human problem, not a technical one. No AI system can make someone care about the safety of other road users. But this system can make the consequences of not caring more immediate, more personalized, and more consistent and that shift, even if modest in any individual instance, compounds across millions of journeys to produce a genuinely safer road environment. That is a contribution worth making.

5. REFERENCES

- [1] World Health Organization, Global Status Report on Road Safety 2023. Geneva: WHO Press, 2023.
- [2] Department of Motor Traffic Sri Lanka, Annual Statistical Report on Motor Traffic 2023. Colombo: DMTSL, 2023.
- [3] Sri Lanka Police Traffic Division, Traffic Accident Statistical Report 2022. Colombo: SL Police, 2023.
- [4] V. Tyagi, S. Kalyanaraman, and R. Krishnapuram, "Vehicular traffic density state estimation based on cumulative road work time," *IEEE Trans. Intell. Transport. Syst.*, vol. 13, no. 3, pp. 1125–1137, Sep. 2012.
- [5] N. Buch, S. A. Velastin, and J. Orwell, "A review of computer vision techniques for the analysis of urban traffic," *IEEE Trans. Intell. Transport. Syst.*, vol. 12, no. 3, pp. 920–939, Sep. 2011.
- [6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proc. IEEE CVPR 2016*, pp. 770–778.
- [7] W. Liu et al., "SSD: Single Shot MultiBox Detector," in *Proc. ECCV 2016*, pp. 21–37.
- [8] G. Jocher, A. Chaurasia, and J. Qiu, "YOLO by Ultralytics (Version 8.0)," Jan. 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>.
- [9] P. Leal, R. Oliveira, A. Borras, and P. Salgado, "A Survey on Multi-Object Tracking Methods," *IEEE Access*, vol. 11, pp. 34205–34234, 2023.
- [10] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple Online and Realtime Tracking," in *Proc. ICIP 2016*, pp. 3464–3468.
- [11] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, "BoT-SORT: Robust Associations Multi-Pedestrian Tracking," *arXiv:2206.14651*, Jun. 2022.
- [12] Z. Zhang et al., "ByteTrack: Multi-Object Tracking by Associating Every Detection Box," in *Proc. ECCV 2022*, pp. 1–21.
- [13] Z. Chen, H. Zhu, and X. Jiang, "Real-Time Traffic Light Violation Detection Using Deep Learning," *IEEE Access*, vol. 8, pp. 174696–174706, 2020.
- [14] R. Singh and A. Gupta, "YOLOv5 and Geometric Polygon-Based Traffic Violation Detection at Signalised Intersections," in *Proc. ICDSML 2022*, pp. 1–6.

- [15] Y. Zhang, K. Liu, and J. Chen, "Multi-Class Vehicle Detection Using YOLOv8 for Intelligent Traffic Surveillance," *Sensors*, vol. 23, no. 11, p. 5089, Jun. 2023.
- [16] D. C. Luvizon, B. T. Nassu, and R. Minetto, "A Video-Based System for Vehicle Speed Measurement in Urban Roadways," *IEEE Trans. Intell. Transport. Syst.*, vol. 18, no. 6, pp. 1393–1404, Jun. 2017.
- [17] B. F. Skinner, *Science and Human Behavior*. New York: Macmillan, 1953.
- [18] P. Diewald, T. Moeller, A. Moeller, L. Roalter, and M. Kranz, "Gamification and Gameful Design for Driver Behavioural Change," in *Proc. EICS 2013*, pp. 1–6.
- [19] L. Wen et al., "UA-DETRAC: A New Benchmark and Protocol for Multi-Object Tracking," *Comput. Vis. Image Underst.*, vol. 193, p. 102897, Apr. 2020.
- [20] F. Yu et al., "BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning," in *Proc. IEEE CVPR 2020*, pp. 2636–2645.
- [21] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *Proc. IEEE CVPR 2016*, pp. 779–788.
- [22] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal Speed and Accuracy of Object Detection," *arXiv:2004.10934*, Apr. 2020.
- [23] G. Jocher et al., "ultralytics/yolov5: v7.0," *Zenodo*, Nov. 2022. doi: 10.5281/zenodo.7347926.
- [24] K. Bernardin and R. Stiefelhagen, "Evaluating Multiple Object Tracking Performance: The CLEAR MOT Metrics," *EURASIP J. Image Video Process.*, vol. 2008, p. 246309, 2008.
- [25] FastAPI. (2024). FastAPI framework documentation. [Online]. Available: <https://fastapi.tiangolo.com>.
- [26] OpenCV Team. (2024). OpenCV 4.8 Documentation. [Online]. Available: <https://docs.opencv.org>.
- [27] React Native. (2024). Official documentation. [Online]. Available: <https://reactnative.dev/docs>.
- [28] Firebase. (2024). Cloud Messaging Documentation. [Online]. Available: <https://firebase.google.com/docs/cloud-messaging>.
- [29] M. J. S. Kumar and A. R. Vignesh, "AI-Based Integrated Traffic Violation Detection," *IJRASET*, vol. 11, no. 7, pp. 1051–1060, 2023.

- [30] R. Kumar and S. Patel, "Advanced Traffic Monitoring and Enforcement Using YOLOv8," IJRPR, vol. 5, no. 4, pp. 211–218, 2024.
- [31] A. A. A. Taamneh, "Road Safety and Traffic Management in Developing Countries," Transport and Telecommunication, vol. 21, no. 2, pp. 115–124, 2020.
- [32] S. Liu et al., "Path Aggregation Network for Instance Segmentation," in Proc. IEEE CVPR 2018, pp. 8759–8768.
- [33] T.-Y. Lin et al., "Feature Pyramid Networks for Object Detection," in Proc. IEEE CVPR 2017, pp. 2117–2125.
- [34] Z. Chen, T. Ellis, and S. A. Velastin, "Vehicle Detection, Tracking and Classification in Urban Traffic," in Proc. IEEE ITSC 2012, pp. 951–956.
- [35] D. Roy, S. Paul, "Detecting Traffic Rule Violations and Promoting Road Safety through Artificial Intelligence," JAIMLNN, vol. 2, no. 10, pp. 99–107, Oct. 2023.
- [36] S. Bhosale and M. Patil, "Smart System for Traffic Violation Detection," IJAR, vol. 10, no. 12, pp. 276–283, 2022.

GLOSSARY

Anchor-free detection: Object detection approach where the model predicts bounding boxes directly without pre-defined anchor templates, as used in YOLOv8.

BoT-SORT: Bootstrapped Object Tracking with SORT a multi-object tracking algorithm combining Kalman filter prediction, Hungarian assignment, camera motion compensation, and visual re-identification.

Bounding box: A rectangle enclosing a detected object in a video frame, described by centre position, width, and height.

ByteTrack: A multi-object tracking technique that uses low-confidence detections in a second matching round to recover partially occluded vehicles an approach incorporated into BoT-SORT.

Camera motion compensation (CMC): A tracking sub-module that estimates and corrects for camera vibration between frames, preventing apparent displacement from being misinterpreted as vehicle movement.

Cooldown timer: A per-vehicle timer that prevents the same weaving episode from generating multiple violation records within a set period after the first confirmation.

Driver Safety Score: A gamified daily score starting at 100 points, decremented when violations are confirmed, and reset at midnight each day.

Evidence package: A structured violation record containing pre-violation, at-violation, and post-violation video frames together with violation metadata.

FastAPI: A high-performance Python web framework used for the TVDS backend REST API.

Firestore Cloud Messaging (FCM): Google's push notification service used to deliver real-time violation alerts to driver mobile devices.

Gamification: Application of scoring, badges, and progress indicators from game design to non-game contexts to motivate behaviour change.

Hungarian algorithm: An assignment algorithm used by BoT-SORT to optimally match new vehicle detections to existing track predictions.

Identity switch: A tracking error where a vehicle is incorrectly assigned the track ID of a different vehicle.

Intersection over Union (IoU): The ratio of overlap area to union area of two bounding boxes standard metric for detection and track matching quality.

IoT (Internet of Things): Physical devices in this system, LEDs and buzzers controlled by an Arduino connected to the digital pipeline to provide real-time physical alerts.

Kalman filter: A recursive estimator used by BoT-SORT to predict each vehicle's position in the next frame based on its current position and velocity.

Lane weaving: A traffic violation characterised by repeated oscillatory lateral crossing of solid lane dividers, inconsistent with a deliberate single lane change.

mAP@0.5: Mean Average Precision at IoU threshold 0.5 the standard accuracy metric for object detection models across all classes.

MOTA: Multi-Object Tracking Accuracy a compound metric accounting for false positives, false negatives, and identity switches together.

NMS (Non-Maximum Suppression): Post-processing step removing redundant overlapping detection boxes, keeping only the highest-confidence prediction per object.

PyTorch: The deep learning framework used for training and running the YOLOv8 model.

Red light violation: The act of a vehicle crossing a stop line at a signalised intersection while the controlling signal displays the red phase.

Rule engine: A software module applying predefined logical conditions to vehicle trajectory and position data to confirm whether a violation has occurred.

SORT: Simple Online and Realtime Tracking the foundational tracking algorithm combining Kalman filtering and Hungarian assignment that BoT-SORT extends.

Trajectory buffer: A ring buffer storing the last 90 frame positions of a tracked vehicle, used by the weaving rule engine to analyse lateral movement history.

YOLOv8: The eighth major version of the You Only Look Once detection family, featuring an anchor-free detection head and C2f bottleneck modules.

APPENDICES

Appendix A - Lane Weaving Detection Logic (Pseudocode)

```
FUNCTION detect_lane_weaving(track_id, trajectory_buffer, lane_lines, current_speed):
    IF current_speed < 15 km/h THEN RETURN No violation

    // Assign lane number to each buffered position
    FOR EACH position in trajectory_buffer:
        assign lane = nearest lane based on x-position relative to lane_lines

    // Detect boundary crossing events
    FOR each pair of consecutive positions:
        IF lane assignment changed THEN record crossing direction (left or right)

    // Apply violation criteria
    IF total crossings < 3 THEN RETURN No violation
    IF crossings are all same direction THEN RETURN No violation (legal lane change)
    IF time since last violation for this track < 10 seconds THEN RETURN No violation

    // All criteria met
    RECORD violation timestamp for this track
    RETURN Violation confirmed, crossing count
END FUNCTION
```

Appendix B - Red Light Detection State Machine (Pseudocode)

```
States per vehicle track: APPROACHING, POTENTIAL_VIOLATOR, CONFIRMED,
PROCEEDING_LAWFULLY
FUNCTION update_red_light_state(track, position, signal_phase):
    IF track.state == APPROACHING:
        IF position inside stop_line_zone AND signal is RED:
            track.state = POTENTIAL_VIOLATOR
        IF position inside stop_line_zone AND signal is GREEN or AMBER:
            track.state = PROCEEDING_LAWFULLY

    IF track.state == POTENTIAL_VIOLATOR:
        IF position enters intersection_zone:
            track.state = CONFIRMED
            capture_11_evidence_frames(track)
            deduct_driver_score(track.plate, 15)
            transmit_violation_record(track)
            trigger_IoT_alert(RED_LIGHT)
        IF signal becomes GREEN before vehicle enters intersection_zone:
            track.state = PROCEEDING_LAWFULLY // Legal — stopped and waited

    IF track.state in (CONFIRMED, PROCEEDING_LAWFULLY):
        IF position leaves monitored zone:
            track.state = APPROACHING // Ready for next cycle
    END FUNCTION
```

Appendix C -YOLOv8m Training Configuration

Hyperparameter	Value
Training epochs	150
Batch size	16
Input image size	640 x 640 pixels
Optimiser	SGD with momentum 0.937
Learning rate schedule	Cosine annealing from 0.01 to 0.001
Warmup epochs	3
Mosaic augmentation	Enabled, probability 1.0
MixUp augmentation	Enabled, probability 0.1
Horizontal flip	Probability 0.5
HSV colour jitter	Hue 0.015, Saturation 0.7, Value 0.4
Random scale jitter	Plus or minus 50 percent
Starting weights	yolov8m.pt (COCO pretrained)
Inference optimisation	FP16 via TensorRT